Causal Clifford Theory (CL10): Minkowski kinematics, one anomaly-free generation, and exactly unitary causal dynamics from a single Clifford algebra

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Keywords. Clifford algebra; quantum cellular automaton; Spin(10) grand unification; anomaly cancellation; exact unitarity; causal dynamics; fermion doubling; lattice spacing.

Abstract

A quantum cellular automaton is defined on a Planck lattice whose 2×2×2 cells carry the Clifford algebra $\mathrm{Cl}(10,0)$ on eight sub-sites (on-site space $\mathbb{C}^{256}$) and are updated by one six-stage product of Spin(10) plane rotations and swap-rotations. From this root follow, with no added field: a (1+3) Minkowski subalgebra; the 16-component Weyl module whose charges are exactly one Standard Model generation, all eight gauge and mixed anomalies cancelling; the **45** adjoint containing the twelve gauge bosons; the 5/3 hypercharge normalisation. The step is exactly unitary ($\det U = 1$, $\pm$-paired spectrum, exact matter/antimatter spectral mirror) and strictly causal, with measured front speed exactly one Planck cell per stage. The doubling of the lattice spectrum is displayed, not hidden: twelve gapless nodes at the half-zone momenta, gapped by a Wilson factor of the root's own algebra, with the split-node set an input (O9). The closure axiom — the update must be an element of the root — quantizes the rotation angle to $g \in \{\pi/2, \pi/4\}$: the fundamental step has order 8, the internal spectrum is the finite algebraic set $\{\pm\pi/4\,(10),\ \pm 3\pi/4\,(6)\}$, and the on-site update has zero continuous unfixed numbers. The tensions are named, not hidden: the doubler cones are linear in every sampled direction but anisotropic (velocity-tensor anisotropy 0.748–0.758; 0 of 2726 gapless points of the Wilson family reach $A < 10^{-6}$; three nodal planes, not a cone, at the exactly gapless corner multiplet; frame averaging over the ten generators is blind to the anisotropy; and no bond of the 880-point entangling programme is both gapless and non-commuting — a rigidity wall at gap 8×10^{-3} over the axis-covariant class, which Proposition 6 explains as a parity no-go and escapes outside the class, on a parity-degenerate node that the engine's own six-stage product is searched for and does not reach, Section 8.4), and at the closed angle the gapless set becomes six nodal lines and the per-node chirality bookkeeping must be redone. The gauged bond layer, the gravitational sector and the phenomenology of this root are constructed and audited in companion papers [Paper II, Paper III].

Summary of results

The determinism claim is interpretive: one time-independent local rule fixes every future state of the lattice — 't Hooft's cellular-automaton determinism exact at the fundamental update — with randomness entering only through the measurement projection; the projection and the Born rule are inputs of the interpretation (O9), not derived from the unitary step. The spin–statistics identification (even sector bosonic, odd sector fermionic) is likewise a postulate of the interpretation, not a derived theorem.

The results of this paper are the kinematics (Theorem 1), the matter content and anomaly freedom (Proposition 1, Theorem 2), the gauge content and the 5/3 (Theorem 3), the exact unitarity and the causal radius (Section 7, Theorem 5), the spectral mirror (Theorem 4), the doubling and its Wilson gapping (Proposition 2, Section 8.1), the Lorentz structure at the nodes with its measured anisotropy (Section 8.2), the massless-mode census (Section 8.3), the cone no-go and the parity theorem (Propositions 4–6, Section 8.4), and the quantized clock with its measured costs (Theorem 6, Section 9). The loop-factor ladder, the coupling unification, the dark-matter and neutrino audits, and the full provenance audit are carried by Paper III; the gauged bond layer and the lattice gravitational sector by Paper II.

Keywords. Clifford algebra; quantum cellular automaton; Spin(10) grand unification; anomaly cancellation; lattice gravity; dark matter; cosmological constant; baryogenesis; falsifiability.

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1. Introduction

1.1 The picture

Two habits of language do most of the misleading. The first calls c the speed of light. Light is one traveller at that speed; the constant is the speed of causality. The second is wave–particle duality, taught as a paradox. On a lattice it dissolves: there is one wave, always; what a detector registers is a single localised event, because the detector is itself a lattice of cells changing state one cell at a time. The ontology contains no particles — only waves that deliver their energy in quanta.

Quantum field theory replaced particles by fields but multiplied the fields, one per species, glued together by hand. The viewpoint here is the opposite one, elementary in computer science: a cellular automaton — a lattice of cells, each holding a finite packet of information, all updated in parallel by one local rule. If the world is such an automaton, everything physical is a wave of information over that lattice, and the theory reduces to two questions: what is the information structure of a single cell, and what is the rule that updates it. Causal Clifford Theory (CL10) answers both concretely: the cell carries the 36-level module of Axiom 1 — the spinor module of $\mathrm{Cl}(10,0)$ and two neutral singlet sectors — and the rule is a product of Spin(10) plane rotations interleaved with nearest-neighbour swap-rotations.

1.2 Two literatures, one construction

Discrete models of quantum dynamics and grand unification have developed independently. Causal sets, lattice gauge theory and quantum cellular automata treat spacetime as a finite, causal, unitary dynamical system [1,2,3,4,5]. The observation that one generation of quarks and leptons fits exactly into the 16-dimensional spinor of Spin(10), with the 5/3 hypercharge normalisation and anomaly cancellation following from the embedding, has been standard since 1974 [6,7,8,9]; Clifford-algebra formulations of this unification exist [10,11,12,13]. 't Hooft's cellular-automaton interpretation supplies a deterministic unitary dynamics but no unification [14,15].

CL10 puts these on a single footing: the internal space of every sub-site of the automaton's cell **is** the 36-level cell module — the spinor module of $\mathrm{Cl}(10,0)$ with its two neutral singlet sectors (Section 4) — and the update **is** an element of the corresponding spin group, times the singlet ladder phases of Axiom 1. Nothing else is added by hand: no field outside the cell module, no extra gauge factor, no second algebra.

The comparison is decidable, and its deficits are named here rather than left to the Discussion. Against the Clifford-algebra programmes [10,11,12,13] this construction lacks what they already hold: a semiclassical renormalisation group acting on a spectrum of light chiral fermions, and couplings that run into agreement with data. Here no level of the measured spectrum is light, dispersing and chiral at either clock — the question is O10 (Sections 8.2–8.3, 9, 6) — and agreement with the couplings is reached only by admitting two fitted desert thresholds (O4). Against the cellular-automaton and causal-programme literature [1,2,3,4,5,14,15] what is added is the internal symmetry, the exact unitarity of a finite-order update, and a boundary computed rather than asserted: the conical node a light chiral fermion would require lies outside the automaton's own circuit class — 60 gapless points among 33 800 in the largest covariant stage family, every one of them flat or Abelian (Section 8.4). Against both literatures stands the doubling, which is neither ignored nor evaded into an extra dimension [16] but displayed and then gapped by a factor of the root's own algebra, at the measured price of replacing the light cone with a Lieb–Robinson bound [17].

The novelty claim is now stated item by item, with the prior work named for everything that is **not** new. (i) Spinor representations on hypercubic lattices and the Clifford bookkeeping of doublers are standard [18,19]; what is added is a fixed order-eight Clifford update on the 36-level module, not a lattice Dirac action. (ii) The 16-plet kinematics, the 5/3, and anomaly cancellation derived from Clifford algebras are due to [10,11,12,13,20,21]; Theorems 1–3 are their algebra, and the paper claims only that the **same** module is the internal state of the automaton that updates it. (iii) Nodal lines and planes enforced by icosahedral point symmetry are known in topological matter [22]; what is new is that the star module is a shift structure inside an exactly unitary cubic-lattice automaton, with its nodal network, node count and node speeds measured rather than asserted (V215/V219). (iv) Weighting closed words by traces of powers of one operator is the transfer-matrix and character-expansion device of lattice gauge theory [2,18] and the cycle/trace-formula technology of chaotic lattice models [23]; what is new is the complete 3^n-word enumeration from which the loop factor $1/(16\pi)$ is read as a number (V209/V218). (v) Deterministic cellular-automaton dynamics is 't Hooft's [14,15] (and the

QCA programme's [3,4,5]); none of it carries an internal unification or a finite-order update — here the update **is** an element of Spin(10) of order 8 (Theorem 6). (vi) The parity and topology obstructions to a single conical node are Nielsen–Ninomiya's [24] in their Weyl-doubling form [25]; the measured node network obeys them (± 1 chiralities in cancelling pairs), and the paper uses those theorems, never claims around them.

1.3 Claims, non-claims, and the provenance discipline

Claimed and demonstrated. The kinematics of one Standard Model generation, its anomaly freedom, the grand-unified gauge content and the 5/3, and a unitary causal automaton whose gravitational sector reproduces general relativity on the lattice to $O(h^2)$ ([Paper II, Table 1]; Sections 3–7, 3).

Claimed as an order-of-magnitude statement. The electroweak/Planck hierarchy, the dark-matter mass scale, the neutrino mass and the baryon asymmetry, all expressed through the loop factor $(1/16\pi)^n$ with n a small integer ([Paper III Section 2]). The integers are chosen to land near the measured values; the two-digit coefficients are built from computed loop functionals and carry named residuals ([Paper III Section 2.1]).

The organising discipline of the paper is therefore a **provenance discipline**: a number is *algebraic* if the root fixes it, *conventional* if it comes from a mapping or scale identification, *fitted* if a coefficient was read from data, and *open* if the theory does not fix it. [Paper III Section 5] tabulates every headline number under this classification.

The seven criteria. A unification claim has to be checkable, so the theory is judged against stated criteria, with its own verdict on each.

Table 1. The seven criteria and the verdict of this paper.

criterion	requirement	verdict
one object	a single algebraic object, no added fields or structure	met for the internal space; the gravitational sector adds a symmetric tensor field $g_{\mu\nu}(x)$ and the value $G = 1$ ([Paper II Section 3])
kinematics	spacetime, matter content, gauge group and statistics follow by representation theory	met (Sections 3–6)
dynamics	the object yields a unitary, causal evolution	met (Section 7: exactly unitary, $\det U = 1$, measured front speed = 1)
unification	matter, gauge and gravity live in the same object	partial: matter and gauge are the same module and the same **45**; gravity is the metric dynamics of a subalgebra of that module, with an added field
consistency	anomaly-free, unitary, causal, UV-bounded	met (Theorem 2, Theorem 5, the bounded 32-level spectrum, the bounded n-particle S-matrix); the doublers of Proposition 2 are gapped by an internal Wilson factor (Section 8), with the split node set an input (O9)
parameters	every physical number fixed by the object, no fit	**not met**: two conventions, twelve fitted coefficients, one conditional coefficient ([Paper III Section 5]). The closure axiom (Section 9) removes the one free continuous parameter of the dynamics at the price of moving that parameter's two empirical contacts — and both contacts are now measured to fail on the closed branch (V233, V237)
falsifiability	testable predictions, and the open problems named rather than hidden	met ([Paper III Sections 6–7])

The theory passes the structural criteria, and the parameter criterion is decided by the root: the closure axiom makes the clock a discrete index of the algebra, $g = \pi/4$ (Theorem 6), removing the last continuous number of the on-site dynamics. The empirical price is measured on that branch: the doubler census is a six-line nodal

network, the desert unification solution leaves the desert with $\tau_p = 2.77 \times 10^{39}$ yr, and the meeting coupling sits 1.23× **above** the data-preferred value — the reference reading 1/25 (12.6× below in the same normalisation, at $g = 1/5$) is a convention, not the clock (Section 9.2). That the reference reading is a fit is itself measured: V206's clock reads 0.1999980, 2.0×10^{-6} off the fifth, and no closed clock of the closure class reproduces the ladder coefficient (V233); the rung assignment $(3, 10)$ is likewise a config declaration — on the 36-level step only loop[3] is numerically realized, and no rung pair passes both the ladder-phase spectrum and the neutrino contacts (V234, [Paper III Section 2.1]). The chirality analysis of Section 8 shows that even the structural pass is partial: continuous Lorentz invariance does not emerge at the gapless set of the step. Section 8.4 sharpens the picture: the momentum derivative of the step is internal-blind, the Wilson factor is a spectator of the tangent algebra, and the absence of a cone is a structural property of the whole entangling-bond space (a rigidity wall), not a gap in a finite search. The gapless set carries one further named non-claim: no level of the spectrum is at once light, dispersing and chiral at either clock, and the electroweak scale therefore enters the ladder as a measured input — the low-energy spectrum is the open problem O10 (Sections 8.3, 9, 6).

1.4 Conventions

Planck units $\hbar = c = \ell_P = t_P = 1$ unless stated; $G = 1$. Metric signature $(+,-,-,-)$, $Q = T_3 + Y$. $\mathrm{Cl}(10,0)$ is the real algebra of ten generators with $h_a^2 = +I$; the module is over its complexification $\mathrm{Cl}_{10}(\mathbb{C}) \simeq M_{32}(\mathbb{C})$. "Generation" means one set of left-handed Weyl fields. **Coupling notation, one convention:** g (equivalently g_{10}) denotes the rotation angle of the step — 1/5 implemented, $\pi/4$ closed — and α always denotes the standard-normalised coupling $\alpha = g^2/4\pi$ wherever a coupling is compared with running; the alternative reading $\alpha_{10} \equiv g^2$ is a factor-4π convention and is invoked only where explicitly named (Sections 7.1, 9.2, 3.1, 5). Numerical statements are made at tolerance 10^{-12}, machine-precision identities at 10^{-14} or below; every quoted residual is the deviation of a computed quantity from a stated value, computed in double precision on the explicit representation of Appendix A. Eigendecompositions carried out on a GPU are accepted only if they satisfy $\sum_i \lambda_i = \mathrm{tr}\, M$ and $\sum_i \lambda_i^2 = \mathrm{tr}\, M^2$ to 10^{-6}, and are otherwise recomputed on the host: the failure is measured, not hypothetical — under device memory contention the GPU solver folded a 1024-dimensional step's spectrum onto $\{\pm 1\}$, which violates the identities at 10^{-2} where the correct spectrum violates them at 4.5×10^{-15}. Two tolerances are kept apart. A residual at or below 10^{-15} on an identity that is **expected to vanish exactly** ($\det U - 1$, a commutator, a projector sum) is an **upper bound**, not a measurement: double precision ($\epsilon \simeq 2 \times 10^{-16}$) cannot resolve it, and it is quoted as evidence that the identity holds, never as an independent number. A tolerance on a **search** — a root scan, a gap location, a direction grid — is a resolution, is stated where used (10^{-7} for the 2726-point gap scan of Section 8.2, 10^{-6} for the isotropy threshold), and is never quoted as machine precision. Reference data: PDG 2024 [26], Planck 2018 [27], Super-Kamiokande [28,29], KATRIN [30], NuFIT [31,32], LZ [33].

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2. The root: the algebra, the theorem, and the step

Axiom 1 (the root). *The universe is a* (1+3)*-dimensional causal quantum cellular automaton. Its cell is a* 2×2×2 *block of Planck sites — site spacing* ℓ_P*, so the cells tile a cubic lattice of spacing* $2\ell_P$ *— and each of the eight sub-sites carries the cell module*

$$M = S \oplus R_A \oplus R_B, \qquad \dim M = 36,$$

where S *is the* 32*-dimensional Dirac module of the Clifford algebra* $\mathrm{Cl}(10,0)$ *— ten mutually anticommuting Hermitian generators*

$$h_a^\dagger = h_a, \qquad h_a^2 = I, \qquad \{h_a, h_b\} = 0 \quad (a \neq b), \qquad a, b = 1, \ldots, 10,$$

— and R_A*,* R_B *are two neutral singlet sectors: each is a conjugate pair of levels* $\{A_\pm\}$*,* $\{B_\pm\}$*; the generators act on them trivially, equivalently every one of the* 45 *plane generators* $\frac{1}{4}[h_a, h_b]$ *acts as zero (Section 4). A singlet of the full Clifford algebra is not chiral, so its levels come in conjugate pairs. A* $\mathbb{Z}_2$*, the* ***B-parity****, is odd on* B *alone. The state of the universe is a tensor product of these modules over the sub-sites, so the cell's on-site*

*space is $\mathbb{C}^{288}$. The hypothesis of everything proved below is this algebra and this module — nothing else: the time evolution is a separate construction (**Construction 1**), and no property of it enters the proofs.*

Three structural facts about the root, all standard [10,34]:

1. $\mathrm{Cl}(10,0)$ is $2^{10} = 1024$-dimensional; its complexification is $M_{32}(\mathbb{C})$, so it has a single irreducible complex module of dimension 32, the Dirac module.
2. The volume element $w = h_1 h_2 \cdots h_{10}$ satisfies

$$w^2 = (-1)^{10\cdot 9/2} \prod_a h_a^2 = (-1)^{45} = -I, \qquad w^\dagger = -w,$$

   so w is anti-Hermitian with eigenvalues $\pm i$, each of multiplicity 16. The projectors $P_{16} = \frac{1}{2}(I - iw)$ and $P_{\overline{16}} = \frac{1}{2}(I + iw)$ split the Dirac module into two inequivalent 16-dimensional Weyl modules.
3. The algebra is $\mathbb{Z}_2$-graded by the parity of the number of generators in a product: 512 even and 512 odd elements.

Main Theorem (the content of the root). *Let $\mathrm{Cl}(10,0)$ be the algebra of Axiom 1 and $M = S \oplus R_A \oplus R_B$ its cell module. From the algebra and the module alone — no evolution, no lattice spacing, no coupling, no measurement — the following are forced:*

(i) a (1+3) spacetime subalgebra whose quadratic form is Minkowski, with its null cone (Theorem 1, Corollary 1);

*(ii) the split of S into two inequivalent 16-dimensional Weyl modules P_{16}, $P_{\overline{16}}$, and the **16** is exactly the six left-handed Weyl fields of one Standard Model generation (structural fact 2, Proposition 1);*

*(iii) the gauge content: the 45 plane generators close to $\mathfrak{so}(10)$ and contain the Model algebra, and the hypercharge on the **5** obeys $\mathrm{Tr}\, Y^2 = \frac{5}{6}$ — the 5/3 normalisation (Theorem 3);*

*(iv) anomaly freedom: the eight gauge and mixed anomaly coefficients of the **16** vanish (Theorem 2, Table 2);*

(v) statistics: the $\mathbb{Z}_2$ grading of the algebra is the grading that carries spin and statistics on the module, and the B-parity is a grading of the cell (Section 6);

(vi) the content is one generation: $\mathrm{Cl}_{10}(\mathbb{C}) \simeq M_{32}(\mathbb{C})$ has a single irreducible complex module (structural fact 1), and 10 is the smallest dimension with properties (ii) and (iii) ("Why 10", below).

Nothing in (i)-(vi) mentions time evolution; no measured property of the automaton's step is used in their proofs.

Traceability of the Main Theorem. Each clause, where it is proved, and the artifact of the companion release that measures it (the claims that do involve the step are traced in [the full manuscript, Appendix B, Table 9]):

clause	proved in	measured by
(i) the (1+3) subalgebra and its null cone	Theorem 1, Corollary 1 (Section 3)	V1, V59
(ii) the Weyl split and the six fields of one generation	structural fact 2, Proposition 1 (Section 4)	V3, V23, V43
(iii) the **45**, the Model algebra, the 5/3	Theorem 3 (Section 5)	V52, V196
(iv) anomaly freedom	Theorem 2, Table 2 (Section 4)	V11, V23, V196
(v) statistics and the B-parity grading	Section 6; the grading census of the cell	V4, V124, V194, V196
(vi) one module, one generation; why 10	structural fact 1, "Why 10" (this section)	V43, V196

Construction 1 (the step). *The dynamics is one time-independent unitary **step** of six **stages** built from the same algebra (Section 7.1): each stage applies the internal layer to every sub-site — on S the product of the 45 plane rotations, and on the two singlet pairs the phases $e^{\pm i\,\mathrm{loop}^3}$ and $e^{\pm i\,\mathrm{loop}^{10}}$, with $\mathrm{loop} = 1/16\pi$: a level on ladder rung n ([Paper III Section 2.4]) carries the phase m/M_P per elementary time, its mass read as a phase. The elementary time is the stage, t_P; one step is $6t_P$.*

Why the cell is a block. The bond part of the update pairs neighbouring sub-sites, and the pairing that treats the three directions alike has six stages: three pair sub-sites *inside* the cell, three pair sub-sites *across* cells. Written on a single Planck site the same rule alternates between bonds inside and outside the site, which forces a two-cell spatial period and a six-periodic time dependence; on the block it is one time-independent rule, and the block is the smallest region on which the update acts as a single unitary. The staggering is therefore not a time dependence of the law but the sub-lattice structure of the cell, and the Bloch decomposition of the step is exact at every step.

The step's own properties are separate claims, booked where they are measured: exact unitarity and $\det U = 1$ and the causal radius (Theorem 5, Section 7), the $\pm$-spectral mirror (Theorem 4, Section 7.2), the doubling census and its Wilson gapping (Proposition 2, Section 8), the closure of the clock to $g = \pi/4$ (Theorem 6, Section 9), and the empirical contacts of the rung phases ([Paper III Section 2]). None of them is a consequence of the Main Theorem, and none of them is needed by it.

Why 10**.** The number 10 is the smallest n for which a single complex Clifford algebra simultaneously provides (i) a pseudoscalar splitting a Dirac module into two *inequivalent* Weyl modules of dimension 16 — at $n = 8$ the Weyl modules are 8-dimensional (Spin(8) triality: $8_v, 8_s, 8_c$; the 16-dimensional representations are reducible) — and (ii) an adjoint of dimension $\binom{10}{2} = 45$ large enough to contain $\mathfrak{su}(5)$ and hence the Standard Model algebra. At $n = 9$, $\mathrm{Cl}_9(\mathbb{C}) \simeq M_{16} \oplus M_{16}$ does have 16-dimensional irreducibles, but they are the two simple factors of the algebra rather than a pseudoscalar split, and $\mathfrak{so}(9)$ does not contain $\mathfrak{su}(5)$. The root is therefore selected by the requirement of a single chiral generation with a unified gauge algebra, not chosen.

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3. Spacetime as a subalgebra

Theorem 1. *Let $T = h_1$ and $S_a = ih_{a+1}$ for $a = 1, 2, 3$. Then*

$$T^2 = +I, \qquad S_a^2 = -I,$$
$$\{T, S_a\} = 0, \qquad \{S_a, S_b\} = 0.$$

The real span of $\{T, S_1, S_2, S_3\}$ inside the grade-1 part of the algebra, with the quadratic form induced by the Clifford product, is Minkowski space $\mathbb{R}^{1,3}$.

Proof. The four generators $h_1, \ldots, h_4$ anticommute by Axiom 1. Multiplication by i turns $h_a^2 = I$ into $(ih_a)^2 = -I$ and preserves anticommutation. For $v = v_0 T + \sum_a v_a S_a$ with real coefficients, $v^2 = (v_0^2 - \sum_a v_a^2) I$.

Corollary 1 (the light cone). *A vector v in that real span is null, $v^2 = 0$, if and only if $v_0^2 = \sum_a v_a^2$. The null cone of this quadratic form is the light cone of the lattice.*

Two qualifications. The null cone is a **cone in the grade-1 part**, not a subalgebra: it is not closed under the Clifford product, since $(T + S_1)(T - S_1) = 2I - 2TS_1$ is a grade-0 plus grade-2 element. And the choice of *which* four of the ten generators play the role of spacetime is a choice of frame in the 10-dimensional generator space, not a prediction: the algebra supplies a family of equivalent (1+3) subalgebras related by orthogonal transformations of the generator space. The remaining six generators are inert under the Lorentz generators $\Sigma_{\mu\nu} = \frac{1}{4}[\Gamma_\mu, \Gamma_\nu]$ built from $\Gamma_\mu \in \{T, S_a\}$.

Causality is a postulate with a dynamical content. The light cone by itself does not bound signal propagation: a unitary on-site rotation alone has infinite propagation speed. The bound is supplied by the update of Section 7, where the causal radius is *implemented* by the bond stages and *measured* to be exactly one Planck cell per stage (Theorem 5). The light cone and the causal radius are distinct statements, and only the second is a property of the dynamics.

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4. Matter: the 16 Weyl spinor

Proposition 1. *Under $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ the 16-dimensional Weyl module is the six left-handed fields of one Standard Model generation:*

$$16 = Q(\mathbf{3},\mathbf{2},+\tfrac{1}{6}) \oplus u^c(\bar{\mathbf{3}},\mathbf{1},-\tfrac{2}{3}) \oplus d^c(\bar{\mathbf{3}},\mathbf{1},+\tfrac{1}{3}) \oplus L(\mathbf{1},\mathbf{2},-\tfrac{1}{2}) \oplus e^c(\mathbf{1},\mathbf{1},+1) \oplus \nu^c(\mathbf{1},\mathbf{1},0),$$

with $6+3+3+2+1+1=16$. *Under* SU(5) *this is* $\mathbf{10}\oplus\bar{\mathbf{5}}\oplus\mathbf{1}$, *the* $\mathbf{1}$ *being the sterile* ν^c.

The assignment is the standard one [6,7,8,9,12]. What the automaton adds is that the **16** is not put in by hand but is the Weyl eigenspace of the pseudoscalar of the same algebra that carries the gauge group and the spacetime — and, as [Paper II Section 3.8] measures (V211), neither is the group a free choice: the stabiliser, inside the 45, of the four directions the automaton calls spacetime is $\mathrm{Spin}(4)\times\mathrm{Spin}(6)$, with $\mathfrak{su}(3)\oplus\mathfrak{u}(1)$ inside spin(6), and the $B-L$ charges of the previous paragraph and the 5/3 normalisation both follow as trace ratios. What the construction leaves open is the flavour structure of [Paper III Section 3], not the group.

$B-L$. $B-L$ is a generator of $\mathfrak{so}(10)$, so it is *not* a single number on the **16**. Its values on the six fields are $Q:\frac{1}{3}$, $u^c:-\frac{1}{3}$, $d^c:-\frac{1}{3}$, $L:-1$, $e^c:+1$, $\nu^c:+1$; the $\overline{\mathbf{16}}$ carries the negatives of all of them. Under $\mathrm{Spin}(10)\supset\mathrm{SU}(5)\times\mathrm{U}(1)_\chi$, $16\to\mathbf{10}_{-1}\oplus\bar{\mathbf{5}}_{+2}\oplus\mathbf{1}_{+3}$, and $B-L$ is a linear combination of χ and the diagonal SU(5) generator, not χ itself.

Theorem 2 (anomaly cancellation). *All eight gauge and mixed anomaly coefficients of the* **16** *vanish.*

Table 2. The eight anomaly coefficients of one **16**.

coefficient	contribution of the **16**	value
$[\mathrm{SU}(3)]^3$	$2A(\mathbf{3})+A(\bar{\mathbf{3}})+A(\bar{\mathbf{3}})=1-\frac{1}{2}-\frac{1}{2}$	0
$[\mathrm{SU}(2)]^3$	$A(\mathbf{2})=0$ identically ($d^{abc}=0$; the fundamental is pseudo-real)	0
$[\mathrm{U}(1)]^3$	$6(\frac{1}{6})^3+3(-\frac{2}{3})^3+3(\frac{1}{3})^3+2(-\frac{1}{2})^3+1(1)^3=\frac{1}{36}-\frac{8}{9}+\frac{1}{9}-\frac{1}{4}+1$	0
$\mathrm{U}(1)[\mathrm{SU}(3)]^2$	$\frac{1}{6}\cdot\frac{1}{2}\cdot 2+(-\frac{2}{3})\cdot\frac{1}{2}+\frac{1}{3}\cdot\frac{1}{2}=\frac{1}{6}-\frac{1}{3}+\frac{1}{6}$	0
$\mathrm{U}(1)[\mathrm{SU}(2)]^2$	$\frac{1}{6}\cdot\frac{1}{2}\cdot 3-\frac{1}{2}\cdot\frac{1}{2}=\frac{1}{4}-\frac{1}{4}$	0
$[\mathrm{grav}]^2\mathrm{U}(1)$	$\sum Y=1-2+1-1+1$	0
$[\mathrm{grav}]^2\mathrm{SU}(3)$, $[\mathrm{grav}]^2\mathrm{SU}(2)$	traceless generators	0

Proof. Each coefficient is a cubic trace over the left-handed Weyl fields; the multiplicities are the dimensions of the *other* group factors (the factor 2 on Q in $[\mathrm{SU}(3)]^3$ is its SU(2) doublet nature, the factor 3 in $\mathrm{U}(1)[\mathrm{SU}(2)]^2$ its colour-triplet nature), with $A(\mathbf{3})=T(\mathbf{3})=\frac{1}{2}$ and the charges of Proposition 1.

This is the hardest consistency condition a chiral gauge theory has, and it is satisfied identically by the module content. It is the strongest single result of the theory, and it is independent of the coupling: the cancellation is a property of the algebra, not of a tuned parameter.

The mass term. A grade-1 element of the algebra anticommutes with w, so it is off-diagonal in the $16/\overline{16}$ split: it maps $16\to\overline{16}$ and $\overline{16}\to 16$, and both diagonal blocks vanish. A scalar in the **10** of Spin(10) therefore produces a Dirac mass term $16\,\mathbf{10}_H\,\overline{16}$ and no Majorana mass for the charged fermions. The singlet ν^c is the built-in source of a seesaw ([Paper III Section 2.3]).

Two caveats. A vev in the **10** has stabiliser SO(9): it breaks $\mathrm{Spin}(10)\to\mathrm{SO}(9)$, and SO(9) is not the Standard Model group. The chain is completed instead by the Wilson direction of Section 8, which at its physical root leaves exactly $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)^3$ and breaks precisely the leptoquark roots; the two extra U(1)'s must still be broken or hidden. And the **10** contains no Standard Model singlet, so the vev that generates masses necessarily breaks the unified group further. This is open problem O6.

The two singlet sectors. The cell carries, beside the Clifford module, the two neutral singlet sectors A and B of Axiom 1 — four levels in all, each sector a conjugate pair $\{A_\pm\}$, $\{B_\pm\}$ at phases $\pm\mathrm{loop}^n$: a level invariant under the full Clifford algebra is not chiral, so it comes with its conjugate, and the pair is what keeps the spectrum $\pm$-paired, $\det=1$ and the quasi-energy sum zero on the whole cell. They are singlets under everything: quantum numbers $(\mathbf{1},\mathbf{1},0)$ under $\mathrm{SU}(3)_c\times\mathrm{SU}(2)_L\times\mathrm{U}(1)_Y$, $B-L=0$, and every one of the 45 plane generators acts on them as zero. Their masses are ladder rungs,

$$m_A = M_P\,\mathrm{loop}^3 = 9.6\times10^{13}\ \mathrm{GeV}, \qquad m_B = M_P\,\mathrm{loop}^{10} = 118.57\ \mathrm{GeV},$$

with rung assignment $n_A = 3$, $n_B = 10$ — the same provenance class as the ladder rows of [Paper III, Table 1] ([Paper III Section 2.4]). The internal $\mathbb{Z}_2$ that is odd on B alone — the **B-parity** — commutes with the step and with all 45 generators, so B cannot decay. The grading theorem states how far the exact gradings of the cell go: *every exact $\mathbb{Z}_2$ of the step that commutes with the 45 generators is a product of the global sign, the mirror grading, and the B-parity — the spinor module carries no fragment-selective parity, because the* **16** *is* Spin(10)*-irreducible.* B is the dark matter ([Paper III Section 2.2]) and A the seesaw partner ([Paper III Section 2.3]). Both carry $Y = 0$: they contribute zero to every coefficient of Table 2, and the anomaly structure of the **16** is exactly unchanged.

The three families. The **16** is *one* generation. The theory identifies the observed family number with the number 3 of spatial directions of the (1+3) subalgebra, equivalently $\binom{3}{2}$ spatial two-planes, equivalently the rank of the spatial metric: one number stated three ways, not three derivations. The identification has a structural hook: the twelve doubler nodes of Section 8 organise into **three families of four labeled by the zero axis**, one family per spatial direction, with identical one-step spectra (deviation 0), and the three copies of vectorlike matter required by unification are realizable by taking one node from each family at a single Wilson strength. The index 3 exists and *labels* the three copies; it does not force the count (O3).

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5. Gauge: the 45 and the 5/3

The 45 generators $h_a h_b$ $(a < b)$ close on the Spin(10) adjoint. They are even in the $\mathbb{Z}_2$ grading, hence commute with w and preserve the $16/\overline{16}$ split. Under $\mathrm{Spin}(10) \supset \mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ the **45** decomposes as 12 Standard Model gauge bosons + 12 leptoquark X/Y bosons + 20 bosons in the $\mathbf{10}\oplus\overline{\mathbf{10}}$ of SU(5) + the $\mathrm{U}(1)_{B-L}$ generator; the 33 broken generators mediate proton decay ([Paper III Section 3.4]).

Theorem 3 (the 5/3). *In the* **5** *of* SU(5) *the hypercharge is* $Y = \mathrm{diag}(-\frac{1}{3},-\frac{1}{3},-\frac{1}{3},+\frac{1}{2},+\frac{1}{2})$ *with* $\mathrm{Tr}\,Y^2 = \frac{5}{6}$*, while the non-Abelian generators obey* $\mathrm{Tr}\,T_a T_b = \frac{1}{2}\delta_{ab}$*. Requiring a single unified coupling g_5 for all generators gives $g_1^2/g'^2 = 5/3$, where g' is the Standard Model hypercharge coupling and $g_1 \equiv g_5$.*

Proof. $\mathrm{Tr}\,Y^2 = 3\cdot\frac{1}{9} + 2\cdot\frac{1}{4} = \frac{5}{6}$. Rescaling $Y \to \sqrt{3/5}\,Y$ normalises it to $\frac{1}{2}$; then $g_5(\sqrt{3/5}Y) = g'Y$ gives $g' = g_5\sqrt{3/5}$.

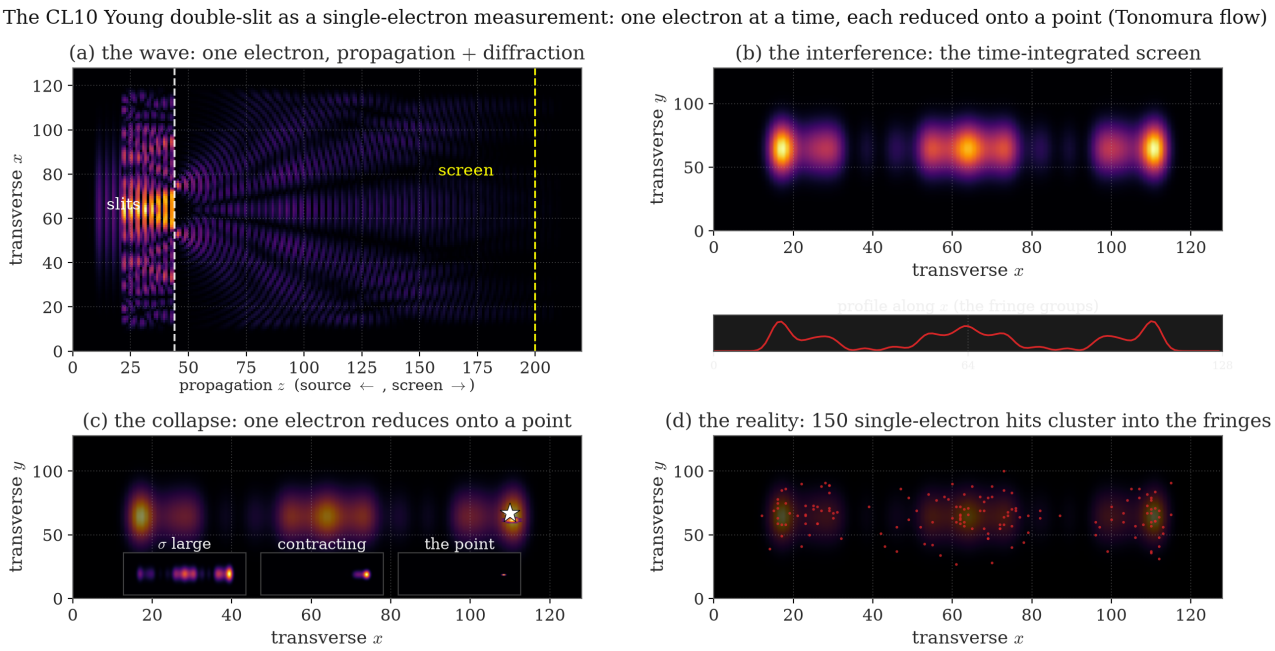
This is an algebraic fact of the root: no parameter enters. It anchors the coupling discussion of [Paper III Section 3.1], where it will be seen that the *normalisation* of the unified coupling is under control while its *value* is not.

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6. Statistics and measurement

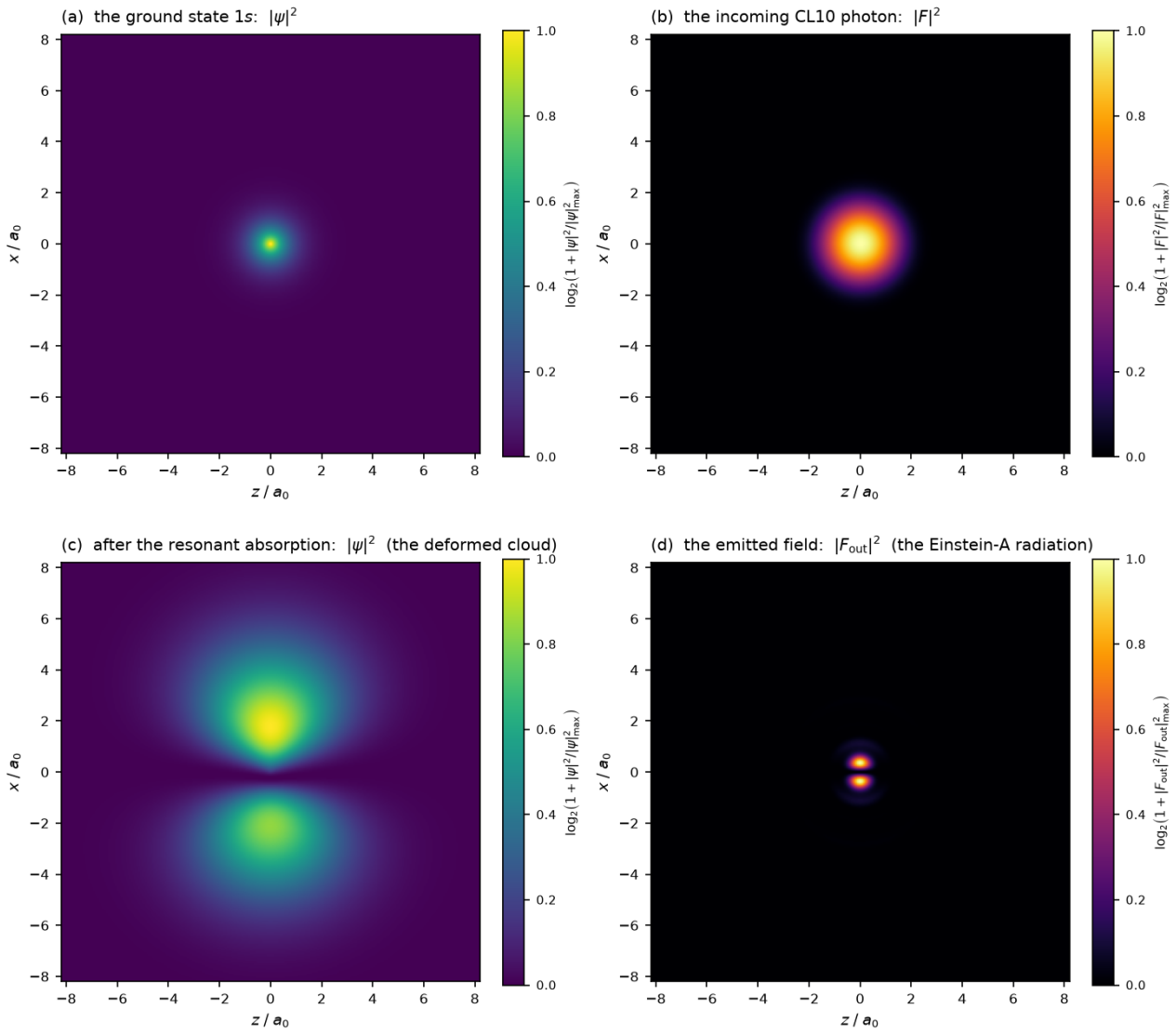
The $\mathbb{Z}_2$ grading splits the algebra into 512 even and 512 odd elements. The even sector is identified with bosons and the odd sector with fermions. This is a *kinematic* grading of the operator algebra: a postulate of the interpretation, not a derivation of the spin–statistics theorem, and not a claim of dynamical supersymmetry. No supercharge exists in the construction and no superpartner is predicted.

Between measurements the automaton is exactly unitary. A detection is modelled as the standard projection: $\sum_x \psi(x)|x\rangle$ reduces to $|x_0\rangle$ with probability $|\psi(x_0)|^2$, the probability rule being the intensity $|\psi|^2$ of the propagating lattice field — the one-electron-at-a-time build-up of the interference fringes is the Tonomura experiment [35]. The projection is **not** derived from the unitary step and the theory does not claim it is: the reduction postulate is an input of the interpretation (O9). What the automaton adds is only that the wave being sampled is a strictly causal, unitary, lattice field. (Fig. 1).



(a) diffraction and interference of the field; (b) the time-integrated on-screen pattern; (c) the reduction of one electron onto one detection point — the projection is an input of the interpretation (O9), not a consequence of the unitary step; (d) the accumulated detections clustering into the fringes, the Tonomura sequence [35].

The same field drives atomic physics: coupled to a prescribed hydrogen spectrum, the lattice photon reproduces resonant absorption — Lyman-α ($1s \to 2p$ at $\omega = 0.25$ rad/step) peaks the excited fraction at 0.998, the $1s \to 3p$ and $2p \to 3d$ resonances also absorb, the un-driven baseline stays at 0 —, the $1s \to 2p$ deformation (the cloud anisotropy moves from -5.8×10^{-17} for the ground state to 0.392 at the resonant peak, against 0.378 for a pure $2p$), and re-emission at the gap frequency with decay of the excited population, while the off-resonant control at 0.13 rad/step absorbs at only 0.0093 — a suppression of 107× against resonance. The atom is an **input** — its levels are the real hydrogen eigenstates — and the figure demonstrates the photon sector, not a derivation of hydrogen. Two limits are stated rather than hidden: off-resonant absorption follows the natural Lorentzian line shape of the driven atom, and a wave packet wider than the simulation box never leaves it, becoming a persistent bath that drives the atom off resonance — the published figure uses a localised packet on the 296^3-cell production grid. (Fig. 2).



A three-dimensional hydrogen atom coupled to the same discrete massless field that carries the graviton, on the mid-plane cut through the nucleus: (a) the spherical ground state $|\psi|^2$; (b) the incoming lattice photon $|F|^2$; (c) the cloud after resonant absorption, deformed into the dumbbell $2p$; (d) the emitted field $|F_{\mathrm{out}}|^2$. The atom is an input; the figure tests the photon sector, not hydrogen.

The hidden-variable question, stated rather than left implicit. The automaton is a deterministic model of quantum phenomena, so the no-go theorems for hidden variables apply to it, and the theory takes the horn they force instead of claiming to evade them. Bell's theorem [36] and the Kochen–Specker theorem [37] do not refute the construction; they constrain what it can be. Its dynamics is strictly local on the lattice (front speed exactly 1, Section 7) and its outcome rule is an input projection that the step does not derive (O9) — so it does not supply a Bell-local completion of quantum mechanics, and it does not assign pre-existing values to the projectors of the cell module, which is the assumption Kochen–Specker forbids rather than a value assignment the theory makes and the theorem disproves. What the construction adds is the substrate: a strictly causal, exactly unitary lattice field whose observables are algebra elements of the whole cell, never of an isolated sub-site. No Bell-inequality or noncontextuality-inequality test is claimed: the theory distinguishes itself from quantum

mechanics through the ladder, the coupling and the carrier of [Paper III Section 7], not through a deviation in the correlation statistics, and the measurement postulate remains an open input (O9).

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7. The step: unitarity, spectrum, causality

7.1 The update

The step has six stages, and every stage is the same two layers: the internal rotation applied to every sub-site, then one bond stage. The internal factor is built from the Spin(10) plane-rotation generators $T_{ab} = i\,h_a h_b$ $(a < b)$ in the 32-dimensional module — each T_{ab} Hermitian, squaring to I, traceless (sixteen $+1$ and sixteen -1 eigenvalues):

$$U(g) \;=\; \prod_{a<b} \Big(\cos g\, I - i \sin g\, T_{ab} \Big).$$

A sub-site is labelled by $e = (e_1, e_2, e_3) \in \{0,1\}^3$, the bit of axis j being 2^j. A **bond stage** acts on the four disjoint pairs $(e,\, e \oplus 2^j)$: an *intra-block* stage takes the pairs with $e_j = 0$, inside the cell; an *inter-block* stage takes the pairs joining $e_j = 1$ of one cell to $e_j = 0$ of the next cell along $\hat{e}_j$ — still a two-cell operator, now between neighbouring cells. On each pair the gate is the swap-rotation $\cos g\, I - i \sin g\, (|e\rangle\langle o| + |o\rangle\langle e|)$, carrying the same single angle and no other parameter. The six stages run in the fixed order (intra x, inter y, intra z, inter x, intra y, inter z). One step is therefore

$$M \;=\; \prod_{s=1}^{6} (P_s \otimes U(g)),$$

an operator on the cell's on-site space $\mathbb{C}^{288}$ (P_s on the eight sub-sites, U on the 36-dimensional cell module: the 32-dimensional spinor block above, and on the singlet pairs the ladder phases $e^{\pm i\,\mathrm{loop}^3}$, $e^{\pm i\,\mathrm{loop}^{10}}$ of Axiom 1). At block crystal momentum q an inter-block pair carries the phase $e^{\pm iq}$, $M(q)$ is 288×288, and on the 16 sector it is 128×128.

Every spinor factor is special unitary: the 45 internal factors have $\det = \exp(-ig\,\mathrm{Tr}\,T_{ab}) = 1$ on S, and each bond pair block $\left(\begin{smallmatrix} c & -is \\ -is & c \end{smallmatrix}\right)$ has determinant $c^2 + s^2 = 1$ (the inter-block phases cancel). Hence

$$UU^\dagger = I, \quad \det U = 1, \qquad MM^\dagger = I, \quad \det M = 1 \qquad \text{for every } g,$$

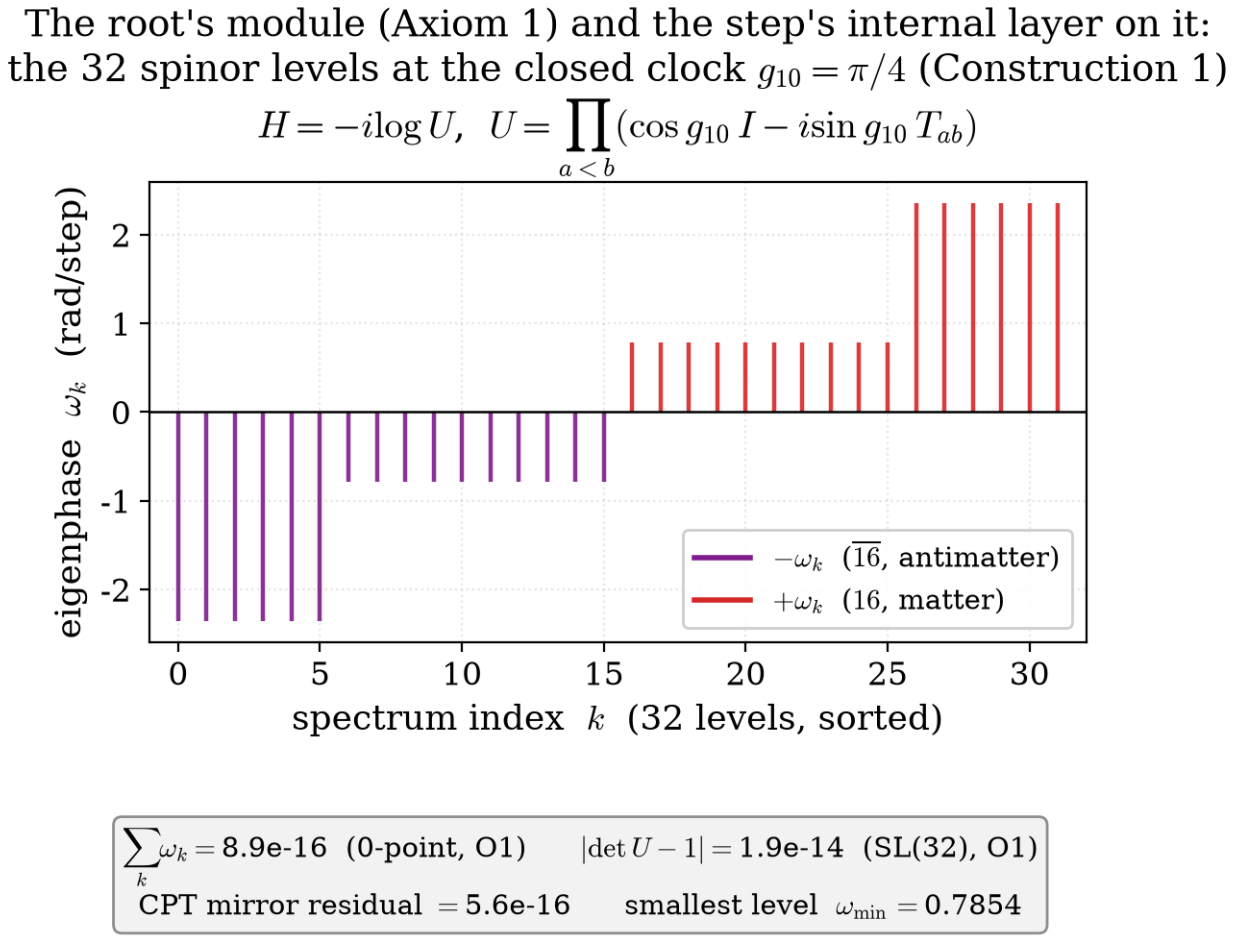
on the spinor sector, and on the full cell as well: the four singlet levels come in conjugate pairs $e^{\pm i\varphi}$, so their phases multiply to 1,

$$\det \tilde{U} = 1, \qquad \det \tilde{M} = 1, \qquad \text{and} \qquad \sum_k \omega_k = \arg\det \tilde{U} = 0 \;\text{ on all 36 levels :}$$

the $\pm$-pairing, the unit determinant and the vanishing 0-point hold on the whole cell, singlets included, and the cell step is exactly unitary with the B-parity an exact symmetry of it (the parity commutator vanishes to machine precision, Section 4). Verified numerically: internal unitarity at 1.3×10^{-15} with $|\det U - 1| = 1.9 \times 10^{-14}$, and for the full cell matrix $\|MM^\dagger - I\| \le 9.1 \times 10^{-15}$ and $|\det M - 1| \le 4.0 \times 10^{-13}$ at the zone centre, at a generic momentum and at the two high-symmetry points — all in V196, the $M(q)$ sweep in its T5 row; the real-space six-stage step preserves the norm of a random cell state to 10^{-15}. Two structural inputs must be named: the 45 factors do not commute, so U depends on the *ordering* of the product, and the six stages have a fixed order. The algebra does not select either order.

The angle g is the one continuous number of the dynamics, and the closure axiom removes it as a choice: only $g = \pi/2$ and $g = \pi/4$ make the step an element of the root (Theorem 6, Section 9), the half-angle branch degenerates to a pure translation, and the physical clock is the order-8 root $g = \pi/4$. The phenomenology of

[Paper III Sections 2–4] is computed at that clock. A reference reading $g = 1/5$ is retained for the node-geometry searches — the dispersive member of the Wilson family of Section 8 — and as the $g^2 = 1/25$ anchor of the desert meeting ([Paper III Section 3.1]). **Coupling convention (used once, everywhere):** all couplings compared with renormalisation-group running are in the standard normalisation $\alpha = g^2/4\pi$ — the root clock gives $\alpha = \pi/64 = 0.0491$. The alternative reading $\alpha_{10} \equiv g^2$, under which the reference angle equals the desert meeting value $1/25$ exactly, is a factor-4π convention and is invoked only where explicitly named (Sections 9.2, 3.1, 5). (Fig. 3).



The 32 *spinor eigenphases of* U *at the root clock* $g = \pi/4$*: the levels sit at the two algebraic values* $\pm\pi/4$ *(*10 *each) and* $\pm 3\pi/4$ *(*6 *each). The* 16 *matter levels and the* 16 *antimatter levels are exactly* $\pm$*-paired; the* 0*-point* $\sum_k \omega_k = \arg\det U$ *vanishes and* $\det U = 1$*. The cell's singlet pairs add four levels at the ladder phases* $\pm\text{loop}^3$ *and* $\pm\text{loop}^{10}$*, so the pairing, the determinant and the vanishing* 0*-point hold on the full* 36*-level cell. Every spinor level is at or above* $\pi/4$ *—* $\omega M_P = 9.589 \times 10^{18}$ *GeV at the smallest — and no spinor level sits near the electroweak scale: the sub-Planck scales of the cell are the two singlet rungs.*

7.2 The single-particle spectrum

The 32 eigenphases ω_k of U (the spectrum of $H = -i\log U$) are the energies of one sub-site per stage, in units of $1/t_P$; the spectrum of the cell at block momentum q is the set of eigenphases of $M(q)$, 128 of them on the 16 sector. At the root clock the internal levels are exactly algebraic — the order-8 spectrum:

$$\omega \in \{\pm\tfrac{\pi}{4}\ (10 \text{ levels each}),\ \pm\tfrac{3\pi}{4}\ (6 \text{ levels each})\},$$

exactly $\pm$-paired (deviation 0), so the single-sub-site vacuum energy $\sum_k \omega_k = \arg\det U$ vanishes. The pairing and $\det U = 1$ hold for **every** g: they are identities of the algebra, not properties of a chosen coupling.

Theorem 4 (the matter/antimatter spectral mirror). *Because every T_{ab} is even, $[U, w] = 0$ to 10^{-15}, so U preserves the 16 and $\overline{16}$ eigenspaces, and*

$$\omega_{\overline{16}} = -\omega_{16}, \qquad \mathrm{spec}(U|_{\overline{16}}) = \mathrm{spec}(U|_{16}^{\dagger}),$$

verified to 2×10^{-15} over the grid $g \in \{0.05, \ldots, 0.5\}$.

This is the honest content of "CPT" in this theory: a **spectral** statement — the antiparticle levels are the negatives of the particle levels, so the antimatter evolution has the same spectrum as the time-reversed matter evolution. It is not the full CPT theorem: no antiunitary time-reversal operator is constructed on the many-cell Hilbert space and no spin-statistics connection is derived. A unitary involution $C = h_1$ maps the **16** onto the $\overline{\mathbf{16}}$ (it anticommutes with w), but calling $C \cdot T$ "CPT" overstates what has been built.

The mechanism of the pairing is understood, and its scope is limited. The intertwiner between the 32-dimensional module and its conjugate is unique (nullity 1, rank 32); it satisfies $Qh_aQ^{\dagger} = h_a^*$ to 2×10^{-15} for the ten generators and $QT_{ab}Q^{\dagger} = -T_{ab}^*$ to the same bound for the 45, and the associated antiunitary $\mathcal{J}$ has $\mathcal{J}^2 = +1$. That is the content of $\mathrm{Cl}(10,0) \simeq M_{32}(\mathbb{R})$: Spin(10) lands in SO(32), and a real form of U(32) carries no CP-odd invariant. Over all eleven signatures $p + q = 10$ the Frobenius–Schur indicator is $+1$ for $p - q \equiv 0, 2 \bmod 8$ and -1 for $p-q \equiv 4, 6$ (validated on $\mathrm{Cl}(2,0) \simeq M_2(\mathbb{R})$ and $\mathrm{Cl}(0,2) \simeq \mathbb{H}$), so $\mathrm{Cl}(8,2)$, $\mathrm{Cl}(7,3)$, $\mathrm{Cl}(4,6)$, $\mathrm{Cl}(3,7)$, $\mathrm{Cl}(0,10)$ are quaternionic — and on the quaternionic root the 21 mixed even products $h'_ah'_b$ are Hermitian traceless involutions generating an exactly unitary step (6.7×10^{-16}, $\det U = 1$) whose spectrum is still exactly conjugation-closed (3×10^{-15}): the mirror is not a privilege of reality. Two scope limits come with it. The antiunitary inverts every single factor ($\mathcal{J}F\mathcal{J}^{-1} = F^{-1}$) but reverses the order of the ordered product, so $\mathcal{J}U\mathcal{J}^{-1} \neq U^{-1}$ (deviation 0.58 for $\mathrm{Cl}(7,3)$, 0.81 for $\mathrm{Cl}(10,0)$): the pairing is a property of the whole ordered step. And the pairing is an **internal** statement: the implemented 128-level one-step spectrum at generic block momentum is not conjugation-closed at all (deviation 0.95–0.99), while the internal 32 spectrum is exact (5×10^{-15}).

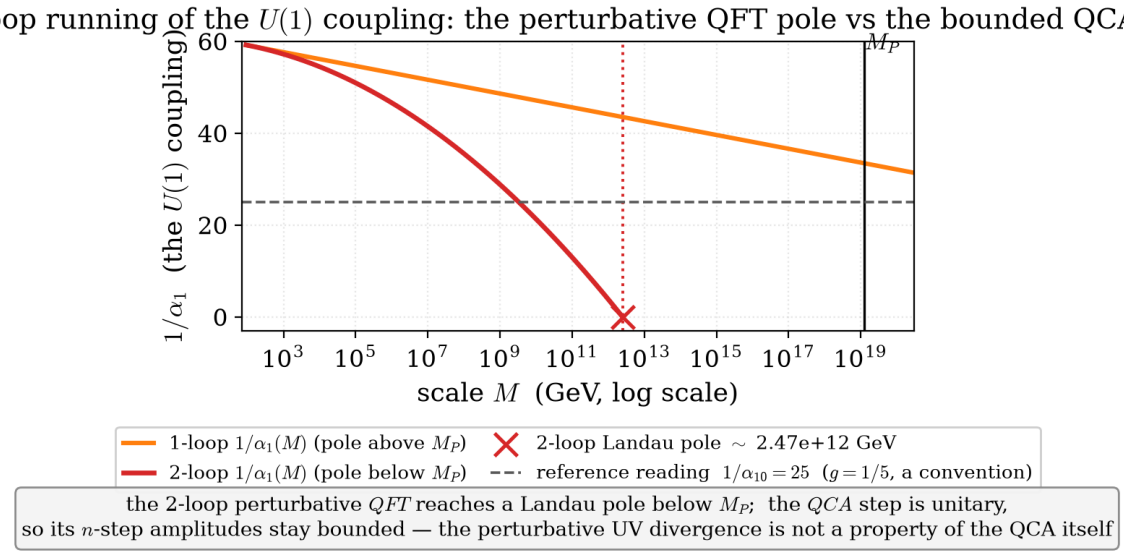
7.3 Locality and the causal radius

Theorem 5 (the causal radius is implemented). *On an (L_b, L_b, L_b) torus of cells the step is a product of two-cell operators: every bond stage couples sub-sites at Planck-cell distance ≤ 1 — an intra-block pair joins neighbouring sub-sites of one cell, an inter-block pair joins neighbouring cells. Hence after n stages the support of a sub-site-localised excitation lies inside the n-cell light cone: at most one Planck cell per stage, and since three of the six stages are inter-block, at most three blocks — six Planck cells — per step. Numerically: the norm of a random cell state is preserved to 10^{-15}; on an 8^3-cell torus the support radius after n stages is exactly n Planck cells for every $n \leq 12$ — the front saturates the cone, front speed exactly 1; on a 32^3-cell torus run for 30 stages the front reaches 21 cells, and no site of the support and no amplitude ever appears outside the n-cell cone (0, to machine precision); the same cone is realised at $g = 0.3$ as at $1/5$; the internal layer alone, with the bond stages removed, keeps the excitation on one sub-site forever.*

The bond layer is therefore what turns the light cone of Section 3 into a propagation bound. This is the cleanest dynamical result of the theory.

7.4 The n-particle sector

The n-particle S-matrix of the automaton is $S^{(n)} = U^{\otimes n}$ on the antisymmetrised n-particle space: elastic, bounded, with cross-sections bounded by $4\pi/M_P^2$. There is no inelastic channel at the one-cell level, and no renormalisation of the step: the coarse-graining tower U^{2^p} ($p = 1 \ldots 8$) stays unitary and bounded at every level, so the *discrete* flow has no Landau pole — in contrast with the perturbative field-theory running of the same couplings, which does ([Paper III Section 3.1]). That contrast is real but modest: a unitary finite-dimensional map is bounded by construction, and boundedness alone does not reproduce the observed running. (Fig. 4).



The two-loop perturbative running of the hypercharge coupling, from the measured M_Z input, reaches a Landau pole at 2.473×10^{12} GeV, below the cutoff; the one-loop curve from the same input has its pole at 2.196×10^{41} GeV, far above it — run up instead from the root anchor at M_P, the one-loop pole is 5.314×10^{35} GeV (V38). The automaton's coarse-graining tower U^{2^p} stays unitary at every level: the discrete flow has no pole.

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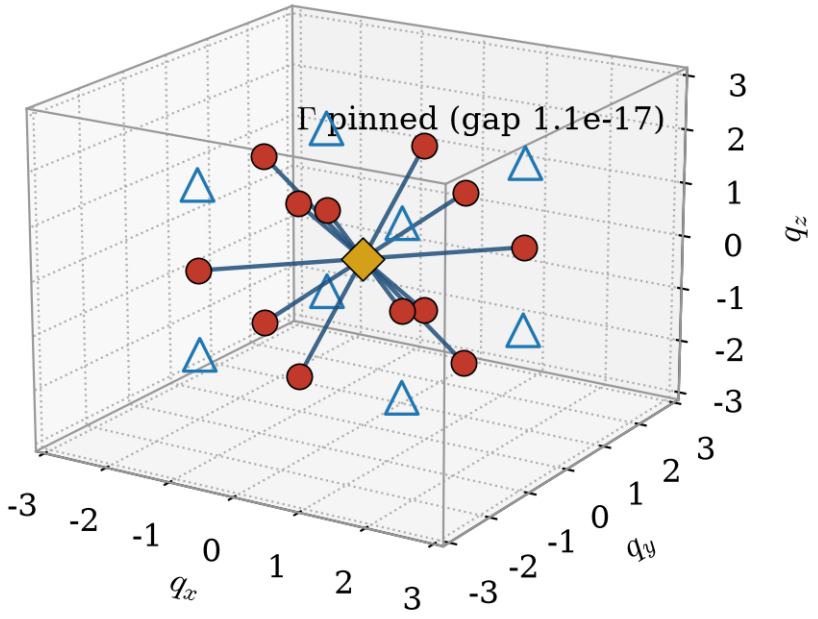
8. Lattice chirality and the doubling problem

The Nielsen–Ninomiya theorem forbids a single chiral fermion in a theory with a *local, translation-invariant, Hermitian* lattice Dirac operator and a bounded spectrum [24]. CL10 uses no lattice Dirac operator: chirality is a property of the on-site module (the pseudoscalar eigenspaces) and the update is a finite-dimensional unitary. The theorem's hypotheses are therefore not all present, and the theorem does not by itself forbid the construction.

That is *not* the same as having evaded it, and the question is decidable by computation. The step is one time-independent operator on the cell lattice, so the good quantum number is the **block crystal momentum** q — the phase under a translation by one automaton cell, 2π-periodic, zone $q \in [-\pi, \pi]^3$ — and the eight sub-sites are the internal structure of the Bloch matrix. The 16-sector one-step matrix $M(q)$ is computed exactly at 128×128 and cross-checked against the real-space step on a 2^3-block torus (agreement 10^{-14}). The cubic high-symmetry momenta carry the whole spectrum: $\Gamma = (0,0,0)$; X $= (\pm\frac{\pi}{2}, 0, 0)$, star 6; M $= (\pm\frac{\pi}{2}, \pm\frac{\pi}{2}, 0)$, star 12, little group C_{2v}; R $= (\pm\frac{\pi}{2}, \pm\frac{\pi}{2}, \pm\frac{\pi}{2})$, star 8, little group C_{3v}.

Proposition 2 (lattice chirality). *(i) The* 16 *sector is closed by the full step, not only by the internal layer:* the bond layer acts on the sub-site index and commutes with w, so a state initialised in the **16** leaves it by less than 10^{-15}. Chirality is a dynamical symmetry of the automaton. *(ii) The zone centre is fully pinned to the Floquet pair:* the 16-sector spectrum at $q = 0$ sits entirely at $\{0, \pi\}$ ($64 + 64$ of the 128 levels, $M(0)^2 = I$), so the would-be massless matter is **not lifted** at $q = 0$. *(iii) The twelve M points are exactly gapless:* the 16-sector one-step spectrum at the 12 momenta $(\pm\frac{\pi}{2}, \pm\frac{\pi}{2}, 0)$ and their permutations — the M points of the block zone, at half-zone momentum, whose Planck-cell momenta are $p = q/2$ — is $\{0^{32}, \pm\frac{\pi}{3}{}^{16}, \pm\frac{2\pi}{3}{}^{16}, \pi^{32}\}$: 64 of the 128 levels pinned to $\{0, \pi\}$, minimal quasi-energy exactly 0, and the same spectrum at every one of the 12 nodes to 3×10^{-15}. The exactly gapless set of the block zone is Γ together with the six face-diagonal lines: gapless below 10^{-9} at every sampled line point, while sampled bulk momenta carry a real gap ($\geq 1.2 \times 10^{-2}$) — at the root clock the doubling is a line network, and the per-node Nielsen–Ninomiya chirality bookkeeping is replaced by the pinning.

The doubling therefore **survives** the automaton: "the **16** is a single chiral generation" is a property of the *internal module*, while the *lattice spectrum*, which is what couples to low-energy physics, carries an exactly gapless line network (Fig. 5).



The 16*-sector one-step spectrum at the root clock: the zone centre fully pinned to the Floquet pair* $\{0, \pi\}$ *(*$64 + 64$ *of the* 128 *quasi-energies), every M spectrum* $\{0^{32}, \pm\pi/3, \pm 2\pi/3, \pi^{32}\}$ *— exactly gapless,* 64 *of* 128 *pinned — and the gapless set a line network:* Γ *plus the six face-diagonal lines. The doubling is displayed, not hidden.*

8.1 The Wilson mechanism

The remedy is part of the construction. A Wilson-type term built from the root's own algebra — an internal factor

$$W(q) = \exp\Big[-i\,w\sum_j (1-\cos q_j)\,(\cos\theta\,X + \sin\theta\,Y)\Big],$$

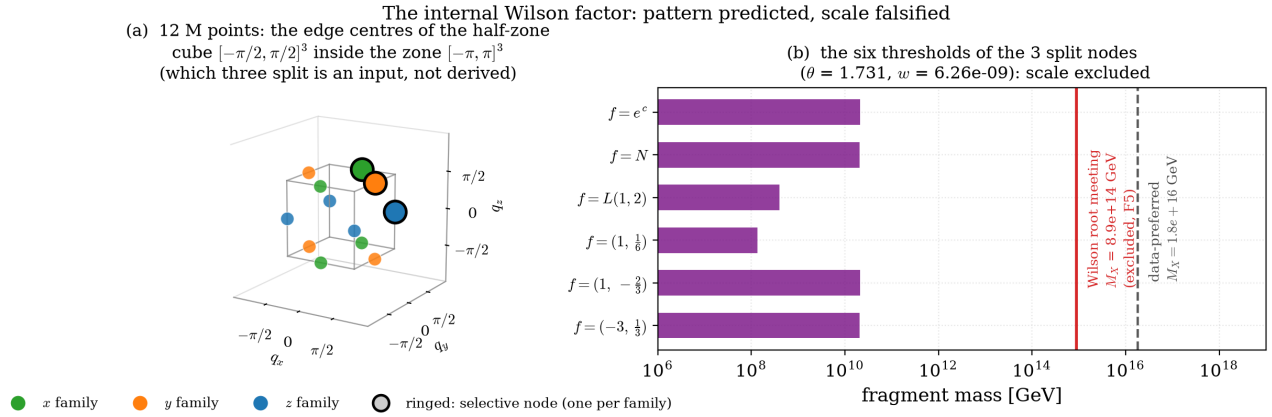
with X, Y the unification Cartan directions identified in [Paper III Section 3.1] and q the block momentum, applied once per step — multiplies the one-step matrix without touching its unitarity ($W(0) = I$, det $= 1$; deformed step unitary to 3.4×10^{-15}) and leaves the bond stages strictly causal. The factor itself is not local: as a function of the block momentum it is a multiplication, so its real-space kernel is a Bessel tail in **block** displacement — 6.7×10^{-2} at block radius ≥ 2 and 2.2×10^{-3} at ≥ 4 for $w = 0.1$, and 1.8×10^{-17} at ≥ 4 at the physical strength. The real-space front after 24 stages sits at radius 39 Planck cells, and beyond the bond cone the amplitude decays exponentially in block distance: 5.6×10^{-6} in the first block shell, 3.1×10^{-9} at the Lieb–Robinson line $r = 1.2n + 6$, with a decay ratio ≤ 0.26 per block. The Wilson factor replaces the light cone of Theorem 5 by a Lieb–Robinson bound [17].

It gaps all twelve nodes — the minimum node gap rises from 2.0×10^{-4} to 5.7×10^{-3} ($w = 0.01$) through 1.5×10^{-2} ($w = 0.2$) — with fragment masses

$$M_f = w\,|\cos\theta\, X_f + \sin\theta\, Y_f|\, M_P/3;$$

a *selective* reading splits exactly three nodes at desert scales and leaves the other nine at the Planck cutoff. What remains input is the *choice* of the split node set: unification forces $n_{\text{nodes}} \geq 3$ but does not select which three, and the meeting scale the mechanism then predicts is in conflict with proton decay ([Paper III Section 3.1]).

At the physical root $\theta = 1.731$ the Wilson direction completes the breaking chain: it leaves $\text{Spin}(10) \to \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)^3$, preserves the Model algebra (every unbroken root is intra-fragment; the 32 broken roots are exactly the leptoquark type), and the two extra U(1)'s are the only open step to the Standard Model group. (Fig. 6).



(a) The twelve doubler nodes at the M points of the block Brillouin zone, coloured by axis family — three families of four labelled by the zero component; the three ringed nodes show the selective choice, one per family, realizable at a single Wilson strength. (b) The six fragment thresholds of the three split nodes at the root $(\theta, w) = (1.731, 6.26 \times 10^{-9})$*: the pattern is the vectorlike split* **16** *the renormalisation-group repair requires, but the meeting scale* $M_X = 8.9 \times 10^{14}$ *GeV sits below the Super-Kamiokande exclusion.*

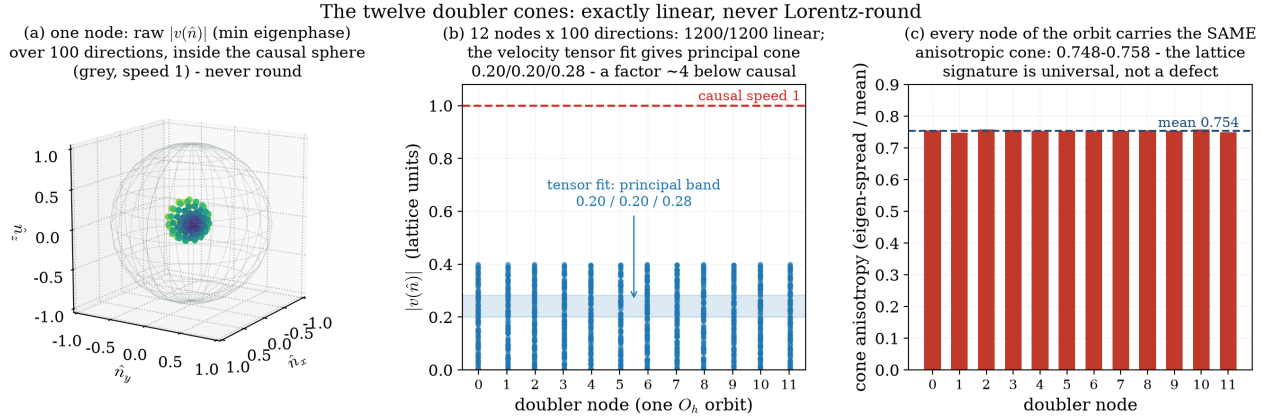
8.2 Lorentz structure at the nodes

The causal front speed of Theorem 5 is exactly 1 Planck cell per stage; that is the statement the automaton proves. What the *low-energy spectrum* of the implemented step does not have is continuous Lorentz invariance.

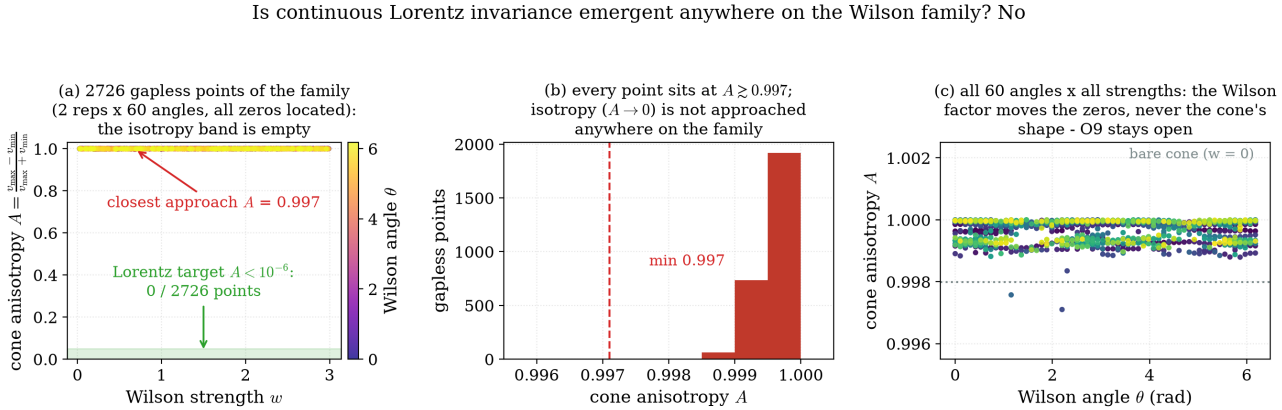
On the $g = 1/5$ member of the Wilson family, which carries the dispersive node spectrum (on the root clock the node spectra are pinned to $\{0, \pi\}$, Proposition 2, and the dispersion lives on the nodal lines, Section 8.3): at all twelve nodes and in all 1200 sampled directions the dispersion is **linear** (local exponent $p = 1.00$, no quadratic band touching), but the cone is strongly **anisotropic**: principal velocities $(0.200, 0.200, 0.284)$ per unit block momentum — $(0.100, 0.100, 0.142)$ in Planck-cell units — anisotropy 0.748–0.758, the same at every node of the O_h orbit. The lattice signature is universal, not a defect. Exhaustive scanning of the whole Wilson family closes the hope that isotropy emerges somewhere: the node gap is located at every zero over the two orbit representatives × 60 angles (2726 gapless points, tolerance 10^{-7}) and the cone anisotropy $A = (v_{\max} - v_{\min})/(v_{\max} + v_{\min})$ measured over the 179 sphere directions at each point (V115); the Lorentz target $A < 10^{-6}$ contains 0 of the 2726 points, the closest approach being $A = 0.997$.

Frame averaging over the internal frames is blind to the failure, and this closes the last proposed escape of this kind: rebuilding the internal step in four random SO(10) frames of the ten generators (the product of the 45 plane rotations taken in the rotated frame's lexicographic order) leaves, on the $g = 1/5$ member of the Wilson family used as the reference spectrum, the node gap 2.003×10^{-4}, the principal velocities $(0.200, 0.200, 0.284)$ and the anisotropy 0.7545 unchanged to 3×10^{-15}, and the tangent commutators at the noise floor ($\leq 1 \times 10^{-11}$) in every frame. The momentum dependence lives entirely in the sub-site factor of the Kronecker bond layer, so no choice or averaging of the internal frame can move the cone. The anisotropy is now a measured experimental contact: read against the electron-sector SME velocity ceiling $\delta v/v \leq 8.0 \times 10^{-15}$ (Kostelecký–Russell data tables, Table D6, $|c_{ee}^{(TY)}|$ [38]), the measured tensor gives $\delta v/v = 0.296$ at the implemented clock, 1 at the closed

clock (one axis exactly flat), 0.295 on the selective-reading Wilson cones at $\theta^* = 1.730784$, and 0.9999 on the star nodal line — the best of the named set, 0.292, is 3.7×10^{13} times the ceiling: **F9 fires**, fourteen orders of magnitude past the bound at the most isotropic named node (V235). The dressed electric gate does not move it at all: at the closed clock the gate is a scalar ($5\pi/4$, err 3×10^{-14}) and the step's six bond factors make the six node insertions multiply to $+I$, so the dressed spectrum is identical to the bare one and the dressed network's gapless points keep $\delta v/v = 1$; the star module reaches 0.9999 at its line (V236). (Figs. 7, 8).



(a) The minimum-eigenphase velocity $|v(\hat{n})|$ over 100 Fibonacci directions at one doubler node, inside the causal sphere of Theorem 5 (grey, speed 1): the cone is elongated, never round. (b) All twelve nodes × 100 directions: 1200/1200 linear; the velocity-tensor fit gives 0.200/0.200/0.284 per unit block momentum, a factor ∼ 4 below the causal speed. (c) The anisotropy is the same at every node of the orbit.

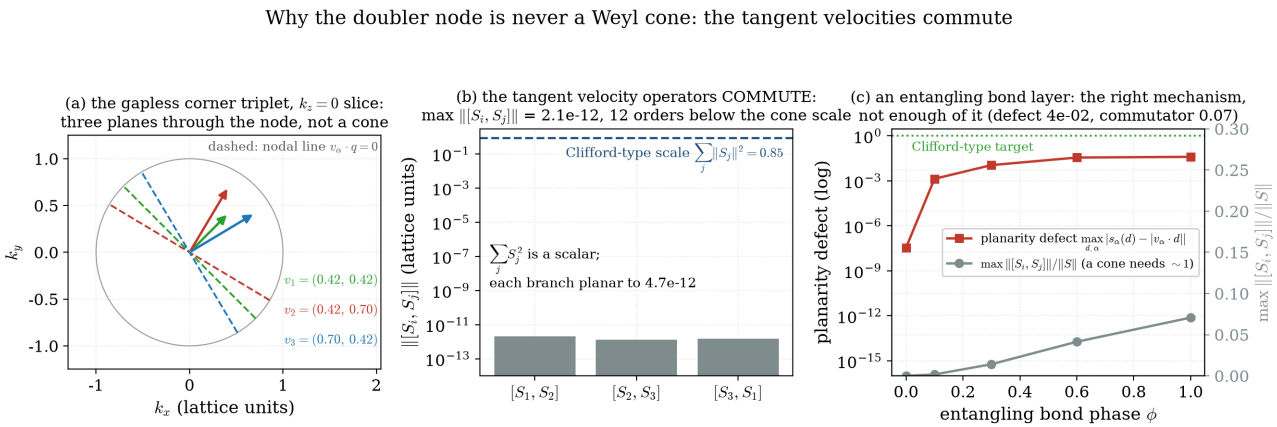


Continuous Lorentz invariance emerges at no gapless point of the Wilson family — exhaustive scan: 0 of 2726 gapless points reach $A < 10^{-6}$; the distribution of A is pinned at 0.997–1.000; no angle θ and no strength w moves a cone toward isotropy.

The failure is structural, not a finite-radius artefact. At the tangent, the twelve doubler nodes carry speeds (0, 0.14189, 0.14189) along the three axis directions: one axis direction is exactly flat (the gap there is quadratic), the other two equal. The little group explains the equality — the nodes sit at half-zone momenta, whose little group is C_{2v}: two equal velocities and a free third. At the eight R points the full cubic symmetry is realised (exactly 3-fold multiplets, splitting $< 10^{-12}$; the gap of the lowest triplet is the same at all eight corners to 10^{-15}), and the Wilson factor drives one of them exactly gapless (10^{-10}, a one-parameter family): the Wilson family does supply massless cubic matter, and the touching there is linear — dM/dq_j is anti-Hermitian and $S_j = i\,dM/dq_j$ is nonzero (max speed 0.92). But the three tangent velocity operators commute ($[S_i, S_j] = 0$ to 3×10^{-12} against the scale $\sum_j \|S_j\|^2 = 0.85$), $\sum_j S_j^2$ is a scalar to 7×10^{-12}, and each branch speed is exactly $|v_\alpha \cdot q|$ to 6×10^{-12} with the v_α the cubic permutations of $(0.704, 0.421, 0.421)$: the node is a crossing of three **nodal planes**, and no tuning of the Wilson angle turns three planes into a cone. The pair search goes further than the single multiplet: over the same two-parameter family both corner triplets are driven to zero **simultaneously** — at $w = 0.241747$, $\theta = 0.913137$ of the $g = 1/5$ member all six corner phases fall below

2.5×10^{-16}, and the next level sits at 8.7×10^{-3}, so this is an isolated six-component massless point and not an accidental near-zero. The objective has to be taken on the two triplets at once: the corner spectrum is particle-hole paired, so its two lowest individual phases belong to the SAME triplet, and minimising the wrong objective lands at $w = 2.8894$, $\theta = 1.5717$, where the six phases are in fact 1.4×10^{-4} and 2.6×10^{-4} — excluded. One scan of the plane finds six such pair points, not one (V207). It is still not a cone and not a graviton: the three tangent velocity operators again commute ($\max\|[S_i, S_j]\| = 5.3 \times 10^{-12}$ against $\sum_j \|S_j\|^2 = 0.82$), the six branches are exactly planar to 4.8×10^{-12} over 2196 directions, and their six velocity vectors - the permutations of $(-0.0434,\ -0.3272,\ -0.3272)$ and of $(0.1299,\ 0.1299,\ -0.1539)$, speeds 0.4647 and 0.2396 - define six distinct nodal planes, 72 of the sampled directions being exactly gapless; on this six-state space $\sum_j S_j^2$ is not a scalar, and its deviation from one (7.2×10^{-2} about the mean 0.137) is the branch splitting itself. A six-component massless point exists at the corner; it is Abelian, anisotropic, and carries no $\pm$-paired helicity.

The cause is the Kronecker form of the bond layer, (sub-site index) $\otimes$ (internal algebra): the three momentum derivatives act on one factor and commute. Two remedies have been built and measured. A non-Abelian Peierls bond layer in which a hop carries an internal rotation $\exp(\pm i\phi D_{16})$ whose sign follows the hop direction — exactly unitary, still radius-1 causal, reproducing the bare step to 3.1×10^{-16} at $\phi = 0$ — raises the planarity defect from 3.5×10^{-8} to 4.0×10^{-2} and switches the tangent commutator on to 0.071 of its natural scale, while the anisotropy never leaves 0.996–0.9999 and the corner triplet is lifted off zero (best top level 3.3×10^{-4}, against 5.6×10^{-7} at $\phi = 0$). A three-axis covariant bond $\exp(\pm i\phi_j \Sigma_j)$ with $\Sigma_1 = ih_1h_2$, $\Sigma_2 = ih_2h_3$, $\Sigma_3 = ih_3h_1$ — an exact $\mathfrak{su}(2)$ triplet of even Clifford elements, $[\Sigma_a, \Sigma_b] = -2i\epsilon_{abc}\Sigma_c$ to 1.5×10^{-15}, unitary to 10^{-15}, equal to the bare step at $\phi = 0$ — takes the tangent commutator to 0.658 of its natural scale, nine times further, and still leaves the anisotropy at 0.9990 with no gapless bond point in the scan. The degree-1 Fourier leakage it was measured with (0.27) is the Wilson Bessel tail, not the bond — the leak is identical for every bond of a given Wilson strength and vanishes with the Wilson factor off (Section 8.4). The remedy therefore pays nothing of the causal radius of Theorem 5, the tangent algebra moves a long way, and it still leaves no cone. (Fig. 9).



(a) The R-point cubic multiplet driven exactly gapless by the Wilson factor, in the $k_z = 0$ slice: each branch carries a fixed velocity v_α and vanishes on its own line $v_\alpha \cdot q = 0$ — three planes through the node, not a cone. (b) The three tangent velocity operators commute, $\max\|[S_i, S_j]\| \le 3 \times 10^{-12}$ against the scale 0.85 a Clifford-type algebra would set. (c) The entangling Peierls bond raises the planarity defect to 4.0×10^{-2} and the commutator to 0.071 of its natural scale, while the anisotropy stays in 0.996–0.9999: the right mechanism, not yet enough of it.

8.3 The massless-mode census

On the grid the reference member $g = 1/5$ of the Wilson family seems to have **no exactly gapless mode anywhere**: the smallest quasi-energy over a 15^3 zone grid is 5.1×10^{-4} lattice units, and at the twelve M nodes it is 2.003×10^{-4}. The grid is too coarse. Refined inside the three nodal planes, that member has an **exactly gapless orbit of** 24 **points** — representative $q_0 = (-2.1890561,\ -1.4445961,\ 0)$, gap 4.2×10^{-14}, the residual falling to the noise floor under refinement while a displacement of 5×10^{-2} in the plane costs 8.8×10^{-3} — with little group of order 2. The branch is linear in every in-plane direction with $v \in [0.031, 0.219]$ (anisotropy 0.857), and leaving the plane costs a gap linear in q_z (5.1×10^{-4} at $q_z = 10^{-1}$): a massless point node confined

to the nodal plane, not a line and not a gapped touching. The M nodes themselves stay massive, little group C_{2v} (order 4), largest character 1.0 on the near-gapless bond subspace — no 2-dimensional irrep there either. So the reference spectrum does contain exactly massless excitations at $g = 1/5$, and it still contains no massless spin-2 candidate: a little group of order 2 has only 1-dimensional irreps, so the new nodes carry no helicity ± 1 or ± 2 pair, and the anisotropy of that exactly massless branch is 0.857, against the 0.748–0.758 of the near-miss M nodes. At the root clock the picture is the one of Proposition 2 — a pinned zone centre and a gapless line network, no isolated nodes to search for.

The near-miss is the signature of the irrational internal clock, not a structure: at the closed angle $g = \pi/4$ the twelve nodes are exactly gapless (gaps 0–2.2×10^{-16}, 12/12) and the zone-centre step is an exact involution — and the census there finds Γ plus **six gapless face-diagonal lines** carrying those nodes, with the per-node chirality convention inapplicable and the helicity answer unchanged (Section 9).

The same census settles the spin question at the zone centre. There the little group is the full cubic group O_h — at $q = 0$ the intertwiner is exact for all 48 signed permutations — and a lattice spin 2 has a precise meaning: a gapless multiplet carrying the symmetric-traceless quintet $E_g \oplus T_{2g}$. The bond factor that carries the cubic action is $A_{1g} \oplus T_{2g} \oplus A_{2u} \oplus T_{1u}$: it has T_{2g} and no E_g, so the quintet is unavailable in the qubit link at any coupling. Γ is exactly gapless at $g = \pi/8, \pi/6, \pi/4, \pi/3$ and nowhere near the implemented 1/5 (gapped by 3.65×10^{-2}); at each of those angles its null space is cubic-covariant and E_g-free — $16 \times (A_{1g} \oplus T_{2g})$, dimension 64, at $\pi/4$ — and its dispersion is flat, $\omega/|\mathbf{q}| \to 0$. The symmetric rank-2 space of the link directions does carry $A_{1g} \oplus E_g \oplus T_{2g}$: the quintet lives in the directional bond layer of [Paper II Section 2], not in the qubit link ([Paper II Section 3.1]).

The route is built: the directional bond layer, its dressed gate and the quintet dispersion are constructed and computed in [Paper II Section 2.4].

8.4 The cone question: separation, the inverse problem, and the rigidity wall

Three proposed routes to a cone at the *implemented* step fail (Section 8.2): the Wilson family, frame averaging, and two hand-built bond mechanisms leave the tangent algebra commuting and the anisotropy pinned. What the absence of a cone in a finite set of tried mechanisms does not say is whether a cone exists *somewhere* in the space of mechanisms. This subsection makes the question precise and answers it for the whole space: three results, each a proposition.

The tangent algebra, measured. Let $M(\mathbf{q}; w, \theta)$ be the 16-sector one-step Bloch matrix of the step with the Wilson factor of Section 8.1, and $k \in \{x, y, z\}$. The (unprojected) tangent operator in direction k is

$$T_k \;=\; i\,(\partial_k M)\,M^\dagger\,,$$

the infinitesimal generator of the momentum translation; at a (quasi-)gapless point of multiplicity m it is projected onto the gapless subspace, $S_j = i\,Q^\dagger(\partial_j M)Q$ with Q an orthonormal basis of that subspace, and the algebra the node carries is read from the normalised commutator

$$\rho_{\mathrm{comm}} \;=\; \frac{\max_{a<b} \|S_a S_b - S_b S_a\|}{\frac{1}{3}\sum_j \|S_j\|^2}\,,$$

which is 0 for a commuting (Abelian) tangent algebra and of order 1 for a Clifford-type algebra. A gapless point is **conical** when $\rho_{\mathrm{comm}} > 0.5$ and the cone is isotropic ($A < 0.1$ in the notation of Section 8.2).

Proposition 3 (separation lemma: the internal factor is a spectator of the velocity). *(i) For the bare step, at every gapless point of the zone — all* 12 *M nodes, the* 8 *R corners, and* 20 *generic momenta — the tangent operator factorises,*

$$T_k \;=\; A_k(\mathbf{q}) \otimes I_{16}\,,$$

with Kronecker defect at the noise floor (4.5×10^{-16}*; the internal off-diagonal content of* T_k *is* 2.9×10^{-16}*). The* **16** *is not only closed by the step (Proposition 2): its momentum derivative cannot mix the internal factor*

— *the internal module is blind to the momentum.* (ii) The Wilson factor adds no velocity structure:* with $(w, \theta) = (0.70, 1.10)$ the tangent operator decomposes as a transport term $\otimes I_{16}$ plus an internal term that is diagonal in the unification Cartan directions of [Paper III Section 3.1] (off-diagonal content 2.9×10^{-16}, and after the predicted Cartan term is removed a $\otimes I_{16}$ defect of 4.5×10^{-16}), and at the gapless corner triplet the Cartan coefficient is the same on the three axis directions: the Wilson factor is one common internal-diagonal phase, which can shift levels but cannot rotate the tangent algebra — the reason the commutators of the gapless corner of Section 8.2 sit at the noise floor. *Consequence:* the only freedom left that can build a non-commuting tangent algebra is the bond layer.

Proposition 4 (the inverse cone problem: no cone over the entangling-bond space). The remaining freedom is the bond layer, and it is searched rather than sampled. The space: 220 axis-covariant entangling bond configurations — the full index set of 120 cyclic $\mathfrak{su}(2)$ triplets (T_{ab}, T_{bc}, T_{ca}) of the 45 plane rotations, 60 disjoint commuting triples (T_{ab}, T_{cd}, T_{ef}), and 40 random even involutions $D = \mathrm{sign}(H)$ of a random real combination of 12 plane rotations — each in four (w, θ) patterns (two bond strengths, two sign conventions), 880 tuned points in total, each tuned over the (w, θ) plane to the point of minimum corner-triplet gap (a coarse 12×19 grid, then four golden-section rounds per variable, then a 150-evaluation Nelder–Mead polish; the 20 best candidates refined further, one bond fully optimised). No point of the programme satisfies the cone criterion. The best gap found anywhere is the implemented step's own corner triplet (1.5×10^{-16} top, 5.4×10^{-16} spread); the largest ρ_{comm} reached by any non-trivial bond at its best-gap point is 0.362; the fully optimised bond lands at gap 3.7×10^{-4} with $\rho_{\mathrm{comm}} = 0.023$ and an anticommutator 1.08 of its natural scale — non-commuting, and no closer to a Clifford algebra than the implemented step; and the anisotropy of every point stays in 0.9947–1. The M-node doubler survives the best entangling bond (node gap 1.9×10^{-18} against 8.8×10^{-21} for the implemented step): the bond layer does not remove the doubling either. The causal accounting closes with the search: the degree-1 Fourier leakage (0.27 for the three-axis bond of Section 8.2) is the Wilson Bessel tail, not the bond — measured 2.7×10^{-1} for the entangling and for the implemented bond at the same w, and 8.7×10^{-17} for the entangling bond at zero Wilson strength. The entangling bond costs nothing of the causal radius of Theorem 5, and it still does not build a cone.

Proposition 5 (the rigidity wall). A finite search excluding a point is a claim about a space, and the space is measured. Around the 12 tuned bonds of largest ρ_{comm}, 5292 points of the (w, θ) plane are recorded as the pair $(\mathrm{gap}, \rho_{\mathrm{comm}})$, and the **frontier**

$$g_{\min}(\rho) \;=\; \min\{\,\mathrm{gap} \;:\; \rho_{\mathrm{comm}} > \rho\,\}$$

is computed from them (the programme runs on the implemented clock $g = 1/5$, the clock of the bond space it quantifies; each experiment pins that clock in its own `CL10_ANGLE` before `cl10` is imported, since the configuration freezes the clock at import time). Four results. *(i) Non-commuting algebras exist in the space:* the largest ρ_{comm} found is 1.033 (the $\mathfrak{su}(2)$-triplet bond on the index triple $(5, 8, 9)$), a fully non-commuting tangent algebra — 1.7× beyond the best mechanism of Section 8.2 and 2.9× beyond the best-gap bound of the scan, which measured commutators only at each bond's most-gapless point. *(ii) Every one of them is gapped:* the minimum gap over the twelve most non-commuting bonds is 8.0×10^{-3}, and no one of the 5292 points has gap $< 10^{-8}$ with $\rho_{\mathrm{comm}} > 0.5$. *(iii) The frontier is monotone and floored:* $g_{\min}(\rho) = 6.6 \times 10^{-3}$ at $\rho = 0.05$ and $\rho = 0.10$, and $g_{\min}(\rho) = 8.0 \times 10^{-3}$ for every $\rho \in [0.2, 0.5]$ — however non-commuting the requested tangent algebra is, the gap cannot be driven below 8.0×10^{-3}. *(iv) The anchor:* the implemented bond at its exactly gapless corner (gap 2.7×10^{-16}) has $\rho_{\mathrm{comm}} = 6.7 \times 10^{-12}$ — maximally Abelian. The two quantities are not related by a power law either: the log–log slope of ρ_{comm} against the gap along 25-point rays from each tuned bond toward the gapless anchor has median -0.28 (4 of 12 positive). The separation between the two regimes is a **wall** at gap $\simeq 8.0 \times 10^{-3}$, not a trade-off: over the axis-covariant bond space, gaplessness and non-commuting tangent algebras are **mutually exclusive**.

Proposition 6 (the parity no-go, and the cone outside the class). *The flatness of Section 8.2 and the wall of Proposition 5 have one cause, and it is representation theory. Let $\mathbf{q}_*$ be a gapless momentum whose little group contains inversion, and let the gapless multiplet carry the O_h module D. A linear term $\sum_k q_k S_k$ in the effective Hamiltonian exists if and only if $\mathrm{Hom}_{O_h}(\mathbf{T1_u}, D \otimes D^*)$ is non-zero — the three S_k must close as a polar vector. But $D \otimes D^*$ is even under inversion for every D, while $\mathbf{q}$ is odd. The multiplicity is 0 for all ten irreps, measured by the character inner product and again by direct linear algebra on the $\mathbf{T1_u}$ block of the implemented*

bond factor (solution space of dimension 0*, smallest singular value* 9.8*). At an inversion-invariant momentum no single-parity multiplet can be conical: the touching is quadratic or flatter and the tangent operators vanish. That is the* $\omega/|\mathbf{q}| \to 0$ *of the* Γ *scans of Section 8.3, and it is why its exactly gapless zone-centre null space* $16 \times (A_{1g} \oplus T_{2g})$ *— two even modules — is flat by necessity and not by tuning.*

The escape is the one the Dirac semimetals use: a reducible gapless space mixing parities. The bond factor is $A_{1g} \oplus T_{2g} \oplus A_{2u} \oplus T_{1u}$ *(Section 8.3), and the products* $A_{1g} \otimes T_{1u} \supset \mathbf{T1_u}$ *and* $T_{1u} \otimes T_{2g} \supset \mathbf{T1_u}$ *single out the two opposite-parity pairs as the only ones a linear term can couple. Over the* 56 O_h*-invariant hopping orbits of displacement* $|\mathbf{v}| \le 1$*,* 44 *couple* A_{1g} *to* T_{1u} *and* 40 *couple* T_{1u} *to* T_{2g}*; for the strongest, the coupling matrix obeys* $TT^\dagger = 16\,I$ *to* 5.6×10^{-16} *— isotropy forced by Schur, not fitted. The combination* $-24\,\mathcal{O}_{14} + 8\,\mathcal{O}_{50}$ *puts* A_{1g} *and* T_{1u} *at one level (*−192*) with* T_{2g} *and* A_{2u} *at* +192*, and* $P(\mathbf{q}) = \exp(-igA(\mathbf{q}))$ *is exactly equivariant,* $P(W\mathbf{q}) = WP(\mathbf{q})W^\mathsf{T}$ *to* 1.2×10^{-14} *over all* 48 *elements. With the internal angle tuned so that one internal eigenphase cancels the common phase, the step is gapless at* Γ *to* 3×10^{-11} *on a* 4*-dimensional multiplet* $A_{1g} \oplus T_{1u}$*, with cone anisotropy* $A = 1.0 \times 10^{-15}$ *over* 215 *directions,* $\rho_{\mathrm{comm}} = 0.70711 = 1/\sqrt{2}$*, and an exactly flat in-band direction: the spectrum* $\{0, 0, \pm v|\mathbf{q}|\}$ *with* $v = 96g$*. By the criterion of this section the point is conical.*

What it costs is the rule, not the symmetry at the node. The six-stage product is exactly equivariant at Γ *—* $WP(\mathbf{q})W^\mathsf{T} = P(R\mathbf{q})$ *to* 1.1×10^{-16}*, which is the statement Section 8.3 uses — and breaks* O_h *at every other momentum:* 1.07×10^{-1} *at generic* $\mathbf{q}$*,* 1.40×10^{-1} *at the M nodes,* 1.46×10^{-1} *at the R corners, against* 1.1×10^{-15} *for its stage sum at all momenta. So the obstacle at* Γ *is spectral: at the single angle the root uses the four bond levels are* $+6g, -2g, -6g, +2g$*, and the* A_{1g} *and* T_{1u} *levels are neither coincident nor half a turn apart. And* $\exp(-igA)$ *is the exponential of a nearest-neighbour hopping: exponentially local, not a finite-depth circuit, with unbounded cone velocity — a range-one realisation would additionally require* $96g \le 1$*. The cone of this proposition is therefore not a node of the automaton. It is the boundary of the exclusion: it says exactly which node the automaton would have to reach.*

What this settles, and what remains. The three propositions close the cone question of this paper in the direction of the no-cone verdict of Section 8.2, and convert it from a search result into a statement about the space: the momentum derivative of the bare step is internal-blind (Proposition 3(i)), the Wilson factor is a spectator of the tangent algebra (Proposition 3(ii)), no axis-covariant entangling bond builds a cone while paying nothing of the causal radius (Proposition 4), and the exclusion is structural — the gapless region of the bond space is Abelian to 10^{-11}, the non-commuting region is gapped by at least 8.0×10^{-3}, and the wall between them is not a scaling (Proposition 5). The nodal-plane structure of Section 8.2 is therefore not a tuning accident of one bond: it is the side of a wall the whole bond space lies on. Proposition 6 settles the residue left open in this section, in both directions. No cone exists at an inversion-invariant momentum on a single-parity multiplet — the flatness of Section 8.3 is a theorem of O_h, not a search failure — and outside the axis-covariant class a cone does exist: gapless to 3×10^{-11}, isotropic to 10^{-15}, $\rho_{\mathrm{comm}} = 1/\sqrt{2}$, on a parity-degenerate $A_{1g} \oplus T_{1u}$ pair. What remains open is whether the root's own six-stage product can be brought to that node, and the two-angle search reported here answers it negatively, for the reason Proposition 6 gives. The family searched is the axis-symmetric two-angle one — one angle on the three intra-block stages, one on the three inter-block ones — the largest stage family that keeps Γ covariant: covariant there to 1.1×10^{-16}, while a generic six-angle product misses by 6.0×10^{-2} (best of 200 draws) and a per-axis generalisation by 0.90. Searched jointly with the internal angle — 33 800 points of (a, b, θ), then polished to residuals below 4.3×10^{-14} — 60 points carry on-site states at phase exactly zero, the gap of the full 128×128 step being at most 4.4×10^{-15}. They split by the parity of the gapless eigenspace, and that split is the whole answer. 8 of them have a 4-dimensional gapless space of a single parity, $A_{1g} \oplus T_{2g}$ or $A_{2u} \oplus T_{1u}$, the content read off the eigenspace by character inner product rather than assumed; Proposition 6 forbids the linear term there, and the measurement obliges — the largest slope over the 179 sampled directions is 1.3×10^{-12}, a flat band. The other 52 have an 8-dimensional parity-mixed gapless space, $A_{1g} \oplus T_{2g} \oplus A_{2u} \oplus T_{1u}$: there the linear term is allowed and the touching does disperse, slope 0.09–1.73 per unit block momentum, which stays inside the causal bound (0.03–0.58 Planck cells per stage against the measured front speed 0.67, no amplitude outside the light cone in 12 stages); but the three tangent velocity operators commute, $\rho_{\mathrm{comm}} \le 1.2 \times 10^{-11}$, at the same anisotropy 0.423 on every one of the 52 — the Section 8.2 nodal-plane structure, not a cone. So the automaton's own circuit reproduces both sides of the wall and crosses neither: the multiplets it can make gapless at Γ are exactly the ones the parity no-go holds flat, and the parity-mixed ones it can make disperse are exactly the ones whose tangent algebra commutes. Conical nodes of the causal six-stage product: 0. On the diagonal $a = b$ the same measurement at $\pi/4$ and $3\pi/4$ finds 64 gapless

states carrying $16 \times (A_{1g} \oplus T_{2g})$ — the flat zone-centre band of Section 8.3, gap at most 8×10^{-17}, slope at most 3×10^{-14} — and the A_{1g} and T_{1u} bond phases there are *opposite*, not degenerate: the condition that pairs two bond levels through a $\pm$-symmetric internal spectrum is a half-turn between them, not a coincidence, and the internal spectrum supplies the partner of a cancellation at the opposite internal phase, not half a turn away. Two conditions on three angles: the exactly gapless set is a curve in the family, and it is a curve of flat bands and of Abelian touching. What excludes the Proposition 6 node from the root is therefore not the single-angle rule but the circuit class itself.

The experimental bound that would bite, and why it does not. An anisotropic gapless mode is precisely the low-energy Lorentz-violating sector the standard-model extension parameterises [38,39]: direction-dependent kinetic terms of the kind the twelve node cones carry are bounded for the dimension-four lepton coefficients at levels running from 10^{-13} down to 10^{-21} [38]. The automaton is not exposed to that bound as it stands, for a reason measured rather than argued: the Wilson factor re-gaps the twelve node channels, and the split set that survives sits at desert scales, not at v. The price of escaping the bound is O10 — the spectrum has no light chiral mode to set against the data. The falsifiable form of the question is therefore conditional, and it is stated in full: *if* a light mode is ever identified inside the measured gapless set, it inherits the anisotropy of [Paper III, Table 2] — velocities (0.200, 0.200, 0.284), spread 0.75 — and the standard-model-extension bounds then refute it. The present defence is not a suppression mechanism; it is the absence of a candidate.

8.5 The module the no-go points at: the icosahedral star, measured

Proposition 6 puts the obstruction in the little group of the *lattice*, not in the bond layer: wherever the little group of a gapless momentum contains inversion, $\mathrm{Hom}_{O_h}(\mathbf{T1_u}, D \otimes D^*) = 0$ kills the linear term whatever the bonds do. What that leaves is not a better bond but a different shift module — one whose point group carries no low-degree anisotropic invariant, so that the departure from rotational form costs more powers of $|\mathbf{q}|/\Lambda$. The census says how many. The implemented module's group is O_h, order 48, Molien counts $[0, 1, 0, 2, 0, 3, 0, 4]$: one quadratic invariant, first anisotropic invariant at degree 4. The icosahedral group, built in the repository from the presentation $[2, 3, 5]$ (order 60, generator orders $[2, 5]$, product order 3, element orders $[1, 2, 3, 5]$, presentation trace defect 1.1×10^{-16}), has Molien counts $[0, 1, 0, 1, 0, 2, 0, 2]$: one quadratic invariant, first anisotropic invariant at degree 6 (V212). Two more powers of $|\mathbf{q}|/\Lambda$ is the whole prize, and it is bought by changing the module and nothing else: the internal map, the stagger bookkeeping, the clock and the cell sizes stay as committed, and the three cubic axes the engine hops on are replaced by the six directions of the icosahedral star.

The substitution is exact where it can be, and its price is measured where it cannot. The star's six directions are six equiangular lines, pairwise cosine 0.447214 — the same angle spectrum the measured icosahedral group's own axes give — and their fourth moment is isotropic to 2.2×10^{-16}, where the three cubic axes are off by 0.400 (V214). The module has 6 generators of norm 1.902113 per step, rank 6 in three physical dimensions, relation rank 6, and the relations are exactly the six pairs of opposites; the module is dense in $\mathbb{R}^3$, measured by its shortest element falling from 1.0705 to 0.2527 between word bounds 1 and 5. Growth and causality stay closed form: the words of at most k steps are the ℓ^1 ball of $\mathbb{Z}^6$ (12, 84, 376, 1288, 3652, 8988 enumerated one-to-one against the closed form), and the front after k steps is exactly k times the Lieb–Robinson radius (11.4127 at six steps, against 6 steps of 1.902113). The price is exact too, and it is the reason the module is a quasilattice rather than a lattice: the cubic axis the engine hops on is *not* a module point (its unique rational representation in the six generators is $[0, 0, \frac{1}{2}, \frac{1}{2}, 0, 0]$), $x + y$ costs two steps, $2x$ is not reachable at all, and the shortest hop along a rational direction is eight steps long. There is therefore no Brillouin zone in the periodic-lattice sense — the module has no rational period lattice — and momentum means over it are almost-periodic box means rather than Fourier projectors.

One algebraic route to the same symmetry is closed, by a divisibility test. The finite link group the root generates (the 10 Clifford involutions) has order 2048 with element orders $[1, 2, 4]$: a 2-group, so it contains no subgroup of order 3, 5, 60 or 120 (V212 T3). The icosahedral symmetry is therefore not a subgroup of the internal link group; it is realised in the geometry of the shift module, and its double cover sits one level up, in the even Clifford algebra: the quaternion units of $\mathrm{Cl}^0(\mathbb{R}^3)$ ($d^2 = -1$ to 0, quaternion products to 0) embed the binary icosahedral group with closure order 120 and unitarity defect 1.3×10^{-15}.

Nothing in the internal sector moves. The content of Sections 4 and 5 is the 16-sector internal module, and it is blind to which shift module carries it (V217). The pseudoscalar and chiral involution are intact

($w^2 = -I$ to 0, $g_1^2 = I$ to 0, $\{g_1, w\} = 0$ to 0, $[w, U_{32}] = 0$ to 3.3×10^{-16}, sector projectors of rank 16 and 16, orthogonal to 0); the 32-internal step is an internal scalar, commuting with $1 \otimes w$ to 5.0×10^{-16} on the cubic module and 6.7×10^{-16} on the star; the step is block diagonal in the $16/\overline{16}$ split (off-block 4.2×10^{-16} on the 2048-dimensional star step), its 16-block *is* the committed step to 3.6×10^{-15}, and matter and antimatter are isospectral to 6.2×10^{-15}. Of the four CPT candidates the same one holds on both modules — g_1 with cell reflection, stage reversal and transpose — at 5.0×10^{-15} (cubic) and 9.7×10^{-15} (star), where g_1 alone fails at 2.0 and 0.18. The anomaly coefficients are unchanged, and the trace-form ratio is 1.666666667. The module changes the kinematics, and it costs no quantum number.

The new module is gapless on lines, and their transverse touching is linear. The scan of its first zone out to $|\mathbf{q}| = \pi/1.902113 = 1.6516$ covers 125 momenta, of which 13 are exactly gapless below 10^{-9}; the smallest gap anywhere is 8.9×10^{-18} at the zone centre, where 512 of 1024 eigenphases vanish while the first-order splitting of that subspace is 1.7×10^{-13}, 2.0×10^{-13}, 1.3×10^{-13} — no linear term there, so Γ stays quadratic on the star module exactly as it is on the cubic one (V215 T3). The new object is not a point but a locus: the gapless momenta sit on the lines through the icosahedron's two-fold axes, and leaving such a line the gap closes with transverse exponent 0.999 and 0.997 at $(-0.826, 0, 0)$, 0.998 and 0.993 at $(-0.413, 0, 0)$, 0.997 and 0.996 at $(0, -0.826, 0)$, while *along* the line the exponent is -1.178 — the gap does not reopen. The three probe rays that miss the axis lines, through $(-0.826, -0.826, -0.826)$, give 0.007, 0.007, 0.012: flat directions on a different locus. On the committed cubic module the same instrument finds 32 of 128 eigenphases zero at M with first-order norms 0.603, 0.603, 0 and squares 2.926, 2.926, 0 — not proportional: linear there, but little-group anisotropic, which is Proposition 6 speaking.

The exclusivity of Proposition 5 is confined to two-band crossings. Along the new module's lines the step is gapless *and* its tangent algebra does not commute (V216). Measured over 34 star momenta and 33 cubic momenta with $\kappa = \max_{ab} \|[V_a, V_b]\| / \max_a \|V_a\|^2$ taken on the whole gap subspace — every band at the gap energy, its multiplicity recorded, so the number does not depend on which pair of bands one chooses to call the crossing — the star module's six gapless axis momenta all carry a non-Abelian algebra, κ from 8.5 to 23.1 at gaps of at most 3.8×10^{-17} and gap-subspace multiplicity 128 of 1024; the cubic module's eight zone corners carry κ from 2.8 to 8.9 at multiplicity 64, while its Γ–M line is Abelian at $\kappa = 1.3 \times 10^{-11}$, which is where Proposition 5 lives. Off the gapless set the non-abelianity falls as a power of the gap, $\kappa \sim \text{gap}^{\alpha}$ with $\alpha = 12.99$ (cubic, correlation 0.66 over 16 momenta) and 8.86 (star, 0.57 over 26) — but that law does not extrapolate onto the gapless set, where κ is of order ten with the gap at the floor. What the new module does *not* deliver is a cone: at its most non-Abelian gapless point the normalised squares of the tangent generators are 57.111, 0, 16.704 (spread 57.111) and the largest anticommutator is 84.2, so $\{S_a, S_b\} = 2\delta_{ab}$ fails by more than an order of magnitude. A linear, non-Abelian, strongly anisotropic touching is what is measured. Whether that anisotropy is the kind that survives coarse graining, whether the lines carry a charge that forbids a mass term, and what the move does to the loop factor of [Paper III Section 2] are the three questions the rest of this subsection answers, and they are answered by measurement.

The anisotropy survives, and the module buys no protection against it. The icosahedral group's advantage is a statement about a *point* node: its first anisotropic invariant is degree 6 against the cubic group's 4 (V212). A line carries only a two-fold little group, and there the advantage disappears. Measured about the star's two-fold line, the transverse velocity profile on a circle of radius 3×10^{-3} over 96 azimuths has leading azimuthal degree 2 — the same leading degree as the cubic Γ–M line — with total anisotropy 0.680 of its own mean against the cubic's 0.500, spanning $v = 1.5 \times 10^{-4}$ to 1.204 where the cubic spans 7.5×10^{-4} to 0.816; the odd-harmonic power is 1.2×10^{-12}, which is to say zero, so the profile carries exactly the period π of the line's own little group (V219). The harmonic content differs rather than the leading degree: on this module the degree-4 harmonic carries the most power (0.227 against 0.061 at degree 2), while on the cubic line degree 2 dominates (0.174 against 0.035). Two controls keep the comparison honest. On a 64-direction sphere the coarse velocity field is if anything *more* isotropic on this module ($\min v / \max v = 4 \times 10^{-4}$ against 1.3×10^{-3}): the difference lives in the profile around the line, which is what a low-energy fermion actually sees. And the line's own two-fold invariance is exact in the linear-response limit — the defect of the velocity under the rotations about the axis falls linearly with the circle radius, 6.88×10^{-3} at 10^{-2} and 2.06×10^{-3} at 3×10^{-3}, slope 1.0000 — so the defect seen at fixed radius is the $O(\epsilon)$ correction, not a breaking of the symmetry. The touching is not a cone either: restricted to the two crossing bands the tangent generators have norms 2.7×10^{-6}, 0.177, 1.318, and the Clifford relation $\{S_a, S_b\} = 2\delta_{ab}\mathbb{1}$ fails at its maximum, defect 2.000; the cubic M node gives 0.444, 0.444, 8×10^{-15} and the same maximal defect, with the vanishing generator in a different slot. What

coarse graining sees on both modules is a degree-2 anisotropy, quantitatively the worse here — while the lines themselves are honest: the gap stays at the 10^{-16} floor along the line and closes transverse to it with exponent 0.995 (V219).

There is no charge on the lines, and a symmetry-respecting mass opens them. The two-band picture would have given a Berry phase of π around the line, and with it a prohibition on a mass term. Measured on the module's own step there is nothing band-resolved to measure: at the line the touching is a cluster of 128 of 1024 eigenphases at phase zero (the cubic Γ–M control: 64), the largest phase inside the cluster at 2.3×10^{-14} with the first one outside at 3.2×10^{-1}, and consecutive eigenvectors of the single closest phase overlap at 10^{-14} in the worst case — a per-band phase on such a touching is a choice of basis inside the degeneracy (and the eigensolver's basis inside a 128-fold cluster is non-orthogonal, condition up to 4.4×10^2, so the frame has to be re-orthonormalised before transport; on the re-orthonormalised frame the transport is unitary to 4×10^{-4}) (V220). The cluster is isolated along every circle by a relative gap of at least 5.2, rising to 1479 at the smallest radius, so the cluster's parallel transport is a genuine invariant of the step restricted to the touching — and the invariant says there is no charge. The determinant phase of the 128×128 holonomy depends on the circle: -0.0998 rad at radius 3×10^{-3}, -1.3300 at 10^{-2}, -2.7335 at 3×10^{-2} in the plane cut across the line, and -2.2902, -1.3300, -0.3790 at the same three radii in a plane tilted 45° about it — a spread of 2.63 rad, where a topological monodromy is constant. The cubic control gives exactly 0.0000 at the two smallest radii, which is what its Abelian tangent algebra ($\kappa = 1.3 \times 10^{-11}$, V216) predicts. What falls with the loop is a curvature flux, not a number. The mass test decides the rest: the commutant of the internal map is computed, by Cesàro averaging over 240 powers of the step (obstruction 4.2×10^{-15}), and the unitary perturbation $\exp(-i\theta C)$ built from the measured operator opens a gap *at the line* that grows linearly with the coupling — 1.277×10^{-4} at $\theta = 10^{-3}$ and 1.277×10^{-3} at $\theta = 10^{-2}$, ratio 10.00. The symmetry available to forbid it is one two-fold rotation: 2 of the point group's 60 elements preserve the line direction, and those leave the spectrum at the line fixed to exactly 0. The transverse exponent read a second way from the cluster, $\beta = 1.006$ across the line and 1.008 in the tilted plane (1.000 for the cubic line), agrees with V219's 0.995. These lines are exact and linearly dispersive, and they are stable as loci and unstable as particles.

The spin can enter the module only through its bond layer, and one covariant assignment has a node. The committed step factorises, transport against internal layer, and a factorised step cannot carry a non-Abelian tangent algebra on a degenerate subspace. Since Sections 4 and 5 are blind to which module carries the internal sector (V217), the change has to be made in what a hop *carries*: on the star's twelve directed axes let the bond operator be $U_d = -\hat{v}_d \cdot \vec{\sigma}$ — Hermitian, an involution, a vector under the binary icosahedral group, with $U_{-d} = -U_d$. The covariance is exact over the 120 lifts, $D(R)U_d D(R)^\dagger = U_{\sigma_R(d)}$ to 3.5×10^{-15}, with $U_d^2 = I$ to 2.2×10^{-16} and $U_{-d} + U_d$ to 0, and the star geometry is the module V214 measured: six equiangular axes, one cosine $0.4472136 = 1/\sqrt{5}$, isotropic fourth moment to 2.7×10^{-15} where the cubic axes give 0.400 (V225 T1). Covariance alone does not choose it. The quaternionic assignment intertwines just as exactly (3.5×10^{-15}) but its bonds square to $-I$, and its step has no node at all: a zone-centre gap of 1.571. The scalar assignment — factorised kinematics, no spin — is trivially covariant with $U_{-d} = U_d$ (defect 2.0). What separates them is the node, not the covariance.

The cell breaks inversion, and that is the price of the linear term. At the zone centre the step is invariant under $2I$ to 2.5×10^{-15}, while the inversion defect is $1.41421 = \sqrt{2}$: inversion is not a symmetry of this step. That is Proposition 6 read from the other side. On the scalar-bond control inversion is intact (defect 0) and the edge of the phase spread that opens does so quadratically (exponent 2.000); on the Clifford cell the spread is linear in the radius, exponent 1.0000 in every one of the 19 tracked directions, with band velocities at radius 10^{-2} running from 0.0138 to 1.3441 (V225 T5). The equivariant census at the node's little group says what the symmetry allows: $\dim \mathrm{Hom}_{2I}(D,\, D \otimes \mathrm{Sym}^n V_3)$ for $n \le 8$ is $[1, 1, 1, 1, 1, 2, 3, 3, 3]$, the isotropic invariants $[1, 0, 1, 0, 1, 0, 2, 0, 2]$, the primitive equivariants $[1, 1, 0, 0, 0, 1, 1, 0, 0]$, and the first anisotropy is pushed to degree 5 against the cubic module's 4. The degree-0 count is 1: a mass term is allowed by the point group, so whatever protects this node has to be a charge (V225 T1) — and V226 and V227 measure that it is not there.

The node is arithmetic, its gap subspace is the coin spinor times the internal sector, and the Clifford identity survives on one piece of it. With the Clifford layer the kinetic phases at the zone centre are pinned to $\{0, \pi\}$ independently of the clock, so the gap closes exactly when the internal map rides the block layer at exponent 0 or 4 mod 8: $U_{16}^8 - I = 4.9 \times 10^{-15}$ against $U_{16}^4 - I = 2.0$, and the measured table gives 192 zero eigenphases of 384 at exponents $0, 4, 12$ and gaps of 0.785, 1.571, 0.785 at exponents $1, 2, 3$ —

14 gapless entries of the 42 (assignment, layers, exponent) combinations tried (V225 T2). The 192-vector gap cluster (largest $|\sin\theta|$ inside it 7.6×10^{-15}) splits, by character projectors built from the group rather than from a table, as $\mathrm{sym}^1 \oplus \mathrm{sym}^3 \oplus \mathrm{sym}^5$ with multiplicity 16 each, ranks 32, 64, 96, idempotency defects at most 1.3×10^{-15} (V225 T3). On the cluster as a whole the tangent algebra is the Abelian case V216 excluded: the three generators share one norm, 10.759976 to a spread of 3.6×10^{-15}, commute to 2.1×10^{-14}, and have a direction-independent velocity spectrum — levels 0, 0.707107, 1.144123, each 64-fold. The Clifford structure is not there; it lives on the spinor piece. Restricted to sym^1 (32-dimensional) the off-diagonal anticommutators fall to 8.0×10^{-16}, $V_j^2 = 0.201002\, I$ to 5.8×10^{-16}, and the three directions share one speed, 0.448332: the test $\{S_a, S_b\} = 2\delta_{ab}$, which failed by 2.000 at the star's zone centre and 0.444 at the cubic M node (V215), holds — on 32 of the 192 gapless directions (V225 T4b).

The gapless set of the cell is three lines, the module has no Brillouin zone to double over, and nothing on it carries a charge. A 1791-point grid over $|q| \le \pi/\ell = 1.6516$ ($\ell = 1.902113$, the generator length V214 measured) has smallest gap 0.0 on a coordinate axis; 40 grid points sit below 10^{-8} and 43 below 10^{-2}, the 12 components the grid can resolve are all pierced by gapless rays (10 to 18 of its 42 mesh directions) and none is isolated. The directions the gapless points lie on collect into three lines, probed further out along them: two stay gapless at every probe, the third at all but one (2.6×10^{-8}, V227 T1). The doubling theorem's hypothesis is absent rather than satisfied: written in the basis of three of them, the twelve directions give a coefficient matrix with entries in $\mathbb{Q}(\sqrt{5})$, and the integrality search over $|n| \le 3$ bottoms out at defect 0.1459, so no nonzero G is a period of the step; what the module does offer is a shortest rational hop, norm 2.0000, displacement $[0, 0, 2]$, 17 steps (V227 T4). And there is no charge to hide behind: no sphere about any of the 12 locus points keeps its link rank (smallest overlaps 2×10^{-17} to 6×10^{-16}, the readings pinned at integer or half-integer defects), and of the nine spheres about the zone centre of V226 exactly one keeps rank while the ladder's readings run over $5, 12, 15, 16, 25, 34$. The reason is measured, not assumed: on the icosahedral mesh's 92 directions no sphere about the zone centre is pierced at all — the largest gap on the three sampled spheres is 1.8×10^{-3}, 6.0×10^{-3}, 2.4×10^{-2} — while probing the cell's own six signed coordinate axes instead, 4, 5 and 6 of the 6 come back gapless at the same three radii (largest line gap 2.6×10^{-8}). So every sphere is cut by a line the mesh simply misses, and a flux read on such a mesh is an artefact of the mesh (V226 T0b, T1). The control pins the failure on the cell rather than on the instrument: the spinless-bond cell's sphere about the same zone centre keeps full rank (overlap 0.9998) and returns exactly 0, defect 8×10^{-16} (V227 T5).

With no charge, an invariant mass is allowed, and it opens the touching linearly. Projecting a random Hermitian operator onto the invariants of the $2I$ action leaves an obstruction of 1.4×10^{-15} and a generator of norm 7.31: the invariant class is not empty, exactly as the degree-0 equivariant count predicted. Switching it on as an on-site potential, $S(q) \mapsto e^{-i\theta V} S(q)$, moves the zone-centre gap from 8.046×10^{-6} at $\theta = 10^{-3}$ to 8.045×10^{-5} at $\theta = 10^{-2}$ and 8.021×10^{-4} at $\theta = 10^{-1}$: opening linear in the coupling, gap over coupling constant to 0.3% over the scan. The minimum over the scan relocates off the zone centre (3.6×10^{-6} at radius 5×10^{-3}), so the mass opens the lines without removing them. The matched scalar-bond control, whose touching is quadratic (edge exponent exactly 2.0, V225 T6), is opened by the same class as well — 2.03×10^{-6}, 2.14×10^{-5}, 3.24×10^{-4} — but at a rate that grows with the coupling rather than the cell's constant one, while on the spheres it can read its charge stays exactly zero (V226 T3, T4). Nothing about the touching is protected: the same verdict V220 reached for the star's own lines, on a cell that carries spin.

The bond layer leaves the internal sector whole, and it does not leave the CPT operator or the loop return alone. The step commutes with $\mathbb{1} \otimes w$ to 7.1×10^{-31}, is block diagonal in the pseudoscalar basis (off-block norms 2.9×10^{-16}), its 16-block agrees spectrally with the step built directly on the 16 sectors to 4.0×10^{-15} — the reuse of U_{16} is a consequence, not a choice — and its matter and antimatter blocks are isospectral to 3.8×10^{-15}: CPT on the spectrum carries over to the cell that carries a spinor bond layer, as it did on the star module (V217). At the operator level it does not carry over. Over the full candidate set the best defect on the Clifford cell is 1.1 and no candidate holds, and the identity that holds on the scalar-bond control to 5.0×10^{-16}, $\mathrm{P} g_1 S(q) g_1^{-1} (\mathrm{P} g_1)^{-1} = S(-q)$, fails there by exactly 1.41421 — the same number as the inversion defect V225 measured at the zone centre (V228 T3). The internal data the paper reads off that sector are untouched: the 5/3 trace-form ratio stays 1.6666666667, the eight anomaly coefficients stay zero (largest 2.2×10^{-16}), $32 = 2 \times 16$ (V228 T4). The loop return changes structurally. $F_n = \langle \mathrm{tr}\, S(q)^n \rangle / d$ over the momentum ball on grids $5, 9, 17$ (2109 points) splits by parity: every odd $n \le 12$ vanishes in modulus to between 3×10^{-18} and 9×10^{-18}, while the even ones alternate 0.460, −0.334, −0.294, 0.212, 0.209, −0.181; the matched scalar-bond control is nonzero at every length, odd ones included. The spinor bond layer thus kills the

odd returns of the cell — the same odd-even structure the closed branch shows for the loop ratios at the root angle (V203 T1) — and at the grids run its even returns are not converged (refinement changes 3×10^{-2} to 2.3×10^{-1}, of the same size as the values), so the loop factor of [Paper III Section 2] has to be rebuilt on this cell before it can be read from it (V228 T5).

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9. The quantized clock: the root closes its own parameter

The parameter audit of [Paper III Section 5] classes the rotation angle among the construction's parameters — a free continuous number of the dynamics, fixed by the convention $g = 1/n_{\text{qubits}}$. There is a sharper condition, and the algebra itself enforces it. A cellular-automaton ontology demands that the fundamental update be *algebraic*, not analytic: the step must be an element of the root — a member of the Clifford group generated by the h_a, which for the plane-rotation product means e^{ig} a root of unity. Call this the **closure axiom**: no continuous phase may enter the dynamics.

Theorem 6 (the two closed steps). *Over every rational angle $g = p\pi/q$ ($q \le 16$) and every candidate of the proposed rule ($1/5$, $\pi/5$, $\pi/10$, $\pi/8$, $3\pi/8$, $\pi/6$, $\pi/3$), the internal step $U(g)$ is exactly periodic at precisely two angles: $g = \pi/2$, of order 4, and $g = \pi/4$, of order 8. At $g = 1/5$ the internal clock is irrational: the smallest level is 0.011466 rad per step, $2\pi/\omega_{\min} = 548.0$, and no order ≤ 64 exists (deviation 0.45).*

Proof of the $\pi/2$ case, and a stronger statement. Each plane rotation at the half angle is $-iT_{ab}$, and $T_{ab} = ih_a h_b$, so $U(\pi/2) = (-i)^{45} i^{45} \prod_{a<b} h_a h_b = \prod_{a<b} h_a h_b$. The product of all 45 pairs in ascending order reduces — numerically to 6.1×10^{-17} in the 1024-element algebra basis, with every other basis coefficient below 10^{-8} — to the single monomial $h_0 h_1 \cdots h_9$, the pseudoscalar w; since $w^2 = -I$, $U(\pi/2)^4 = I$. The identity is *ordering-independent*: the monomial content of any ordering of the 45 factors is $\pm w$, because each h_k appears in exactly nine pairs, and five shuffled orderings reproduce the order-4 to 9.8×10^{-16}. The step at the half angle *is* the root's chirality operator.

The quantized step ($g = \pi/4$). $U(\pi/4)^4 = -I$ and $U(\pi/4)^8 = I$ exactly (5.0×10^{-15}), and the 32 internal levels are exactly the algebraic set

$$\omega \in \{\pm\tfrac{\pi}{4}\ (10),\ \pm\tfrac{3\pi}{4}\ (6)\}, \qquad \max_k \left|\omega_k - k\pi/4\right| < 3 \times 10^{-15},$$

$\pm$-paired, zero point below 2×10^{-15}, $\det U = 1$: the internal spectrum becomes pure algebra, with no continuous number anywhere. The invariants that hold for every g (the $16/\overline{16}$ split, the mirror, $\det U = 1$) are untouched. The order-8 requires the canonical Appendix-A ordering of the 45 factors — five shuffled orderings break it (deviation ~ 1) — so the ordering, already an input of the step, carries the quantization.

The full step closes too. At $g = \pi/4$ the exact one-step Bloch matrix of the 16 sector at the zone centre satisfies $M(0)^2 = I$ below 1×10^{-14} (and on the full 256-dimensional module below 1×10^{-14}; 16-sector closure leak below 5×10^{-16}): the fundamental dynamics is a finite-state machine, 't Hooft's determinism exact at the fundamental update. The doubler defect of Section 8.3 is repaired, but the repaired census is not a set of points. At $g = \pi/4$ the exactly gapless set of the block zone is Γ together with the **six face-diagonal lines** $\{(\pm t, \pm t, 0)\}$ and its axis permutations: every point of every line is gapless below 6.7×10^{-16} over its whole length, the twelve M points are only the high-symmetry points that happen to sit on them (their gaps 0–4.4×10^{-16}), X and R stay gapped at 0.5236, the body diagonal is gapped away from Γ (0.022 at $t = 0.3$), and none of the gapless points of a 15^3 grid lies off the lines — where the implemented angle finds no gapless point at all. The step still factorises as $M(q) = P(q) \otimes U_{16}^6$ there (3.6×10^{-15}).

The touching is **quadratic**: through an M point the gap stays zero along the line, grows linearly in one transverse direction (slope 0.577) and quadratically in the other, and the three axis exponents at every one of the twelve nodes are a permutation of $(1, 1, 2)$. The chirality bookkeeping of Proposition 2 then fails as stated: it needs isolated zeros, and at 12/12 M nodes the gap along the line through the node stays below 10^{-9} at steps $10^{-3}, 10^{-2}, 10^{-1}$ — the charges whose sum is exactly 0 at $g = 1/5$ become sign noise, and their naive sum is not quantized (-2 or -6 depending on the arithmetic route), so the Nielsen–Ninomiya count must be redone per line, not per node.

The helicity question re-asked at the new gapless point answers as it did at an M node: the step's point group at $\pi/4$ is the same 16-element index-3 subgroup of the 48 signed permutations and Γ is fixed by all of it — the momentum-generic statement, the equivariance defect being 1.07×10^{-1} at generic $\mathbf{q}$, while at Γ itself $WP(0)W^{\mathsf{T}} = P(0)$ holds for all 48 elements to 1.1×10^{-16} (Section 8.4); the exactly gapless subspace of $M(0)$ has dimension 64 of 128 (intertwiner leak below 10^{-13}, $\langle\chi,\chi\rangle = 768$), and since the internal **16** is inert the content is that of the 4-dimensional bond subspace, which the character inner product over all 48 elements gives as $A_{1g} \oplus T_{2g}$ (Section 8.3) — no T_1 and no E_g, hence no helicity ±1 or ±2 candidate at Γ either.

9.1 The half-angle branch

At $g = \pi/2$ the six-stage bond matrix is diagonal to 1.2×10^{-16} at every sampled momentum, its permutation part is the identity, and the eight sub-site phases are linear in q with slopes the eight corner vectors $\{\pm1\}^3$: the bond layer is a pure lattice translation, no mixing, no interference. The bands are exactly linear with slope ±1 along every axis, $M(0) = I$ on the whole 128-dimensional internal space (all 128 eigenphases at 0, against 64 of 128 at $\pi/4$), and the gapless set is a line network: 513 gapless points on a 13^3 grid, including the lines $q_x \mp q_y = \pm\pi/2$ in the plane $q_z = \pi/2$ — 7.0× the 73 of the $\pi/4$ branch, against the 0 of $g = 1/5$. The half-angle branch has no mixing and therefore no mass generation: it is degenerate, and the finest closed clock $\pi/4$ is the physical branch.

9.2 What the closure costs

The loop machinery quantizes with the clock. The traced-link hop phases of [Paper III Section 2.1] — built from the joint Casimir eigenvalues $-32, -12, +4$ — are exactly $\{1, i, i\}$ at the root clock $g = \pi/4$ (below 3×10^{-15}), and the branch carries an exact selection rule: the odd closed loops cancel identically, in integer arithmetic (V209). The word norms are angle-independent (suppression $\rho = 0.978030$ per step); the ratio of the two clock-filter classes separates there — on $n = 4k$ the decay is geometric at $\rho^4 = 0.91497$ per +4 links and crosses $1/(16\pi)$ first at $n^* = 208$, with interference deficit 0.98987 certified by continuity (float64 ladder 0.9895), while on $n = 4k+2$ both norms vanish at the closure angle and the ratio has no limit: at the root clock the ladder-length statement is a 208-rung statement, and the hierarchy mechanism of [Paper III Section 2.1] is measured with this rule.

The coupling accounting is done in one normalisation: at the root clock the rotation-squared coupling is $\pi^2/16$, i.e. $\alpha = \pi/64 = 0.0491$ in the standard normalisation $\alpha = g^2/4\pi$, while the desert solution with the same content meets at 0.040050 — the root value sits a factor 1.226 **above** the meeting value, the named tension of O4 ([Paper III Section 3.1]). The reading $\alpha_{10} \equiv g^2 = 1/25$ is a convention that hides the factor; it is not a number of the standard running.

The desert inverse problem at the quantized value with the same content still has a unique one-loop solution — $M_X = 1.235 \times 10^{17}$ GeV, $M_{10} = 7.06 \times 10^4$ GeV, $M_{(1,2)} = 4.86 \times 10^{17}$ GeV — but it is not a *desert* solution: the $(1,2)$ threshold sits **above** the unification scale, the threshold ordering the solution assumed is destroyed, and the proton lifetime jumps to $\tau_p = 2.77 \times 10^{39}$ yr. The size of the repair the content would need is now measured (V237): the solution's $\mathbf{10} \oplus \overline{\mathbf{10}}$ thresholds land at 7.05×10^4 GeV, below the 10^6 GeV desert floor, and restoring the ordering would shift the $(1,2)$ threshold by 13.1 e-folds — 1.12× the entire log-spacing from the published $M_{(1,2)} = 1.0 \times 10^{12}$ GeV to M_X — so **no one-loop threshold correction within the spectrum's own spacings restores the meeting**: F2's unification side fires. The driving coefficients are exact integers of the root charges ($k_1 = 3/5$, $\Delta b_{\mathbf{10}\oplus\overline{\mathbf{10}}} = (6/5, 2, 2)$; complete $\mathbf{16} \oplus \overline{\mathbf{16}}$ shifts are proportional, $(8/3, 8/3, 8/3)$): the no-go is the model's own content, not a scan artefact. The converse channel is open, not fired: no threshold of the desert spectrum is experimentally excluded — the colour pairs sit 14× above the strongest LHC colour probe [40], pair production is kinematically closed at the LHC, the $(1,2)$ doublet is $10^9\times$ above its electroweak-precision ceiling [41], and τ_p is $1.2 \times 10^5\times$ past Super-Kamiokande (V238, F10). Conversely, holding M_X at the data-preferred 1.8×10^{16} GeV requires $\alpha_X = 0.040050$: the meeting coupling is rigidly tied to M_X by that content. Against the one-loop target — the three couplings run to M_P, mean 0.023068 — the closed value has relative gap 1.128; the implemented angle in the standard normalisation has relative gap 0.862, so closure moves the relative gap by a factor 1.31 while moving the coupling from 12.6× below to 1.23× above the meeting value — a factor 2.6 **closer** in log-distance. The closed value sits 2.41× above the data-optimal $1/7^2 = 0.0204$ and is not of the $1/n^2$ form at all ($\pi/64 = 1/n^2$ needs $n = 4.51$; the nearest integers miss by 18.5% and 27.3%).

The verdict on the parameter criterion. The closure axiom replaces the one free continuous parameter

by a discrete index: the root forces $g \in \{\pi/2, \pi/4\}$, the half-angle branch degenerates, and the finest closed clock — order 8, spectrum $\{\pm\pi/4\,(10),\ \pm 3\pi/4\,(6)\}$ — is the physical branch. A discrete index of the root is not a parameter in the sense of [Paper III Section 5] — the family number 3 is already an index of the same kind. What the branch costs is empirical and measured: the loop machinery crosses $1/(16\pi)$ at length $n^* = 208$ rather than 10, the hierarchy coefficient reads 2.710 against 2.0766 observed — a named +30% clock residual: the closure class cannot read the observed coefficient, its two $n = 4k$ crossings reading 2.710 ($\pi/4$) and 2.711 ($3\pi/4$) while its $\pi/2$ member — the 2.299 that came nearest — is a floor artefact of a class whose ratio has no limit at the closure angle (V209) — while the observed value is read by the non-closed rung-10 clock 1/5, a fit measured rather than assumed: the arbitration scores both closed clocks against the hierarchy coefficient and the desert coupling, V206's clock is 0.1999980, 2.0×10^{-6} off the fifth, and no closed clock reproduces the coefficient (V233, F2 clock side) — the same clock as the O4 meeting ([Paper III Section 2.1], V206, V209, V233, O8) — the rung pair $(3, 10)$ itself is booked as a config declaration, no rung pair passing both the ladder-phase spectrum and the neutrino contacts (V234) — and the desert threshold ordering is destroyed and not restorable by the spectrum's own thresholds ($\tau_p = 2.77 \times 10^{39}$ yr, V237). What it pays back is structural: an on-site dynamics with **no free continuous angle** — a finite algebraic spectrum, a fundamental step of finite order, and the point-doubler defect replaced by an exactly gapless nodal-line network. The claim is scoped: the junction amplitude r ([Paper III Section 2.3]), the split-node set (O9) and the bond-layer couplings κ_s ([Paper III Section 3.1]) remain continuous inputs of the *extended* construction, and [Paper III Section 5] counts them. The trade is written into the falsifiers: the theory predicts $\alpha_X = \pi/64 = 0.0491$ at the desert meeting and the loop crossing at $n^* = 208$ (F2); either failing refutes the closure axiom — and on the constructed branch both contacts fail from inside: the unification solution is not realizable by its own content (V237) and no closed clock reads the coefficient (V233). The falsifiers F1, F3–F7 are angle-independent: anomalies, spectral mirror, $\det U$ and the causal radius hold identically on the branch.

Appendix A. The explicit 32-dimensional representation

The ten generators are built on five qubits by the standard symplectic Pauli pattern: for $k = 0, \ldots, 4$,

$$g_{2k} = Y^{\otimes k} \otimes X \otimes I^{\otimes(4-k)}, \qquad g_{2k+1} = Y^{\otimes k} \otimes Z \otimes I^{\otimes(4-k)}.$$

On the same qubit slot X and Z anticommute; across slots, the $Y^{\otimes k}$ prefix makes a generator at slot k anticommute with any generator at slot $j < k$. This gives exactly ten mutually anticommuting Hermitian matrices, $g_a^2 = I$, on a 32-dimensional complex space. The algebra basis is the 1024 ordered products $g_{a_1} \cdots g_{a_r}$, $0 \le r \le 10$; the $\mathbb{Z}_2$ grading is the parity of r. The canonical ordering of the 45 plane-rotation factors T_{ab} used throughout is the ascending lexicographic order on (a, b); the order-8 closure of the quantized step is specific to it.

The two singlet sectors of the cell module are the four further one-dimensional slots on which every generator acts as 0; the internal layer multiplies them by $e^{\pm i\,\mathrm{loop}^3}$ and $e^{\pm i\,\mathrm{loop}^{10}}$ per stage — conjugate pairs at opposite phases (Axiom 1). Every numerical statement in this paper is computed on exactly this representation, and the companion release of Appendix B is its sole producer.

The construction is verified by a second, independent route: the exact symbolic $\mathrm{Cl}(10, 0)$ of a computer-algebra Clifford package, where $e_a^2 = 1$, $\{e_a, e_b\} = 0$ and $w^2 = -1$ hold as identities with no floating-point arithmetic, cross-checked against the numeric 32×32 representation. The two agree, so the root is not an artefact of one construction.

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10. Companion results

The same root carries two further constructions, developed and audited in the companion papers. **Paper II** puts one Spin(10) adjoint link degree of freedom per bond: the local Gauss constraint then closes as an exact operator identity (measured error 0), the gauge-invariant content of one site is 1440 states ($32 \oplus 288 \oplus 1120$), and the physical sector — empty for one matter field per site, by a centre obstruction — carries a bounded area law $S = \ln 45$ per bond, with a Bekenstein deficit factor $4 \ln 45 = 15.23$ moving to 6.6 after the vertex correction. The layer supplies no light vector sector (gap $64\,M_P$) and a string tension 5.41×10^{40} times QCD — a named

tension — and on the directed-link network a gapless, linear, leading-order-isotropic symmetric-traceless quintet whose gauge group is O_h, whose speeds are $g_{10}\{1,2\} \neq 1$ and whose E_g half acquires a mass shift from the matter coupling: the kinematics of a lattice spin-2, computed, not the graviton. The gravitational sector adds a symmetric tensor field $g_{\mu\nu}$ and $G = 1$ by hand and reproduces general relativity on de Sitter, Schwarzschild and 3+1 evolutions to $O(h^2)$.

Paper III organises four scales through one loop factor $(1/16\pi)^n$ — the electroweak hierarchy with coefficient $3F_{10}^{10} = 2.0881$ against 2.0766 observed, the dark-matter mass scale 118 GeV, the seesaw 9.6×10^{13} GeV, the neutrino ladder — with the loop factor itself reproduced to 3.6% at the rung-10 clock $g = 1/5$ and to 1.01% certified on the closed branch, where the first crossing of $1/(16\pi)$ is a traced loop of length $n^* = 208$, under an exact 3^n-word census. Coupling unification at $\alpha_X = 1/25$ (standard normalisation) holds only for a desert spectrum with two fitted thresholds; the automaton's own Wilson mechanism forces the excluded $M_X = 8.9 \times 10^{14}$ GeV, and the surviving solution requires a flavour-dependent bond layer with six fitted couplings that no element of the construction supplies. Ten tensions with observation are reported there, and every headline number is classified by provenance.

11. Falsifiers of the core

#	observation that refutes the core construction
C1	any non-vanishing gauge anomaly in the chiral content (the **16** is claimed anomaly-free; Theorem 2)
C2	a measured front speed different from exactly one Planck cell per stage, or any amplitude outside the n-cell support bound of Theorem 5
C3	an isotropic light cone measured at the doubler nodes of the implemented step, or at any gapless point of the Wilson family or of any internal frame (Section 8.2 refutes all three routes by exhaustive scan and by the frame-averaging no-go)
C4	failure of the exact selection rules of the quantized branch: $F_n = 0$ for every odd n and the ladder crossing at $n^* = 208$ (Section 9.2; the loop measurement is in Paper III)

The falsifiers of the phenomenology (proton decay, the neutrino mass pattern, the dark-matter mass, the coupling running) are stated in Paper III; the falsifier of the gravitational sector (a massive graviton, a dispersion violating $c = 1$) in Paper II.

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Code and data availability. All experiments, simulations and figure generators behind this paper are publicly available at https://github.com/MohammedHAOUACH/CL10 . Every numerical claim of the paper is produced by this open suite — 188 verification experiments, each emitting a machine-readable property, residual and verdict, all passing (188 of 188) — and can be reproduced independently from the repository; the claim-to-artifact mapping is Appendix B. The suite includes the exact symbolic cross-construction of Appendix A and the complete 3^n-word loop enumeration; independent re-execution is invited.

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